Quantum Magic in the Kitaev Model

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Abstract

This note aims to clarify the physical meaning of the roles the matter and gauge sectors play in the quantum magic. We compute the *stabilizer Rényi entropy* (SRE) of the Kitaev ground state with and without the magnetic field by exploiting the spin-string structure of the Kitaev model. The result shows that the gauge sector acts as a stabilizer, and the magic resource comes completely from the matter sector.

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1 Stabilizer Rényi Entropy of the Kitaev Ground State

The **Kitaev ground state is a stabilizer state** due to the Wilson symmetry, and the **stabilizer group is constituted by the plaquette operators** $\langle\{W_p\}\rangle$. The stabilizer projector can also be understood as a *condensation operator* proposed in Ref. [1], which is related to the non-invertible symmetry in a two-dimensional system. The ground state density matrix reads [2]

$$|\mathrm{Gnd}\rangle\langle\mathrm{Gnd}| = \frac{1}{2^{N-N_p}}\left(\frac{\mathbb{1}+\tilde{W}_v}{2}\right)\left(\frac{\mathbb{1}+\tilde{W}_h}{2}\right)\prod_p\left(\frac{\mathbb{1}+W_p}{2}\right)\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}$$

where $\mathfrak{C}_{\mathfrak{D}} \equiv \eta_{\mathfrak{D}}\langle\Sigma_{\mathfrak{D}}\rangle$. Since a pure-state density matrix is a projector, the ground state $|\mathrm{Gnd}\rangle$ can be obtained by acting the density matrix on an arbitrary state $|\phi\rangle$

$$|\mathrm{Gnd}\rangle = C(\phi)\underbrace{\left(\frac{\mathbb{1}+\tilde{W}_v}{2}\right)\left(\frac{\mathbb{1}+\tilde{W}_h}{2}\right)\prod_p\left(\frac{\mathbb{1}+W_p}{2}\right)}_{\text{gauge (=stabilizer)}}\underbrace{\left[\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}\right]}_{\text{matter}}|\phi\rangle \tag{1.1}$$

where $C(\phi) \equiv 2^{N_p-N}\langle\mathrm{Gnd}|\phi\rangle^{-1}$ is a normalization coefficient depending on the initial state. The above form of a state is reminiscent of a state in a *stabilizer space*. A stabilizer space is a subspace in which the states are the eigenvectors of the elements in a *stabilizer group* with eigenvalues $+1$. Note that if the dimension of a stabilizer space is one, we call the only state in such a stabilizer space a *stabilizer state*. The generators $\{g_i\}$, which are commuting $\mathbb{Z}_2$ operators, generate a *stabilizer group* denoted by $\langle\{g_i\}\rangle$. The projector that projects states onto the stabilizer space $\mathcal{V}_S$ is

$$\hat{\Pi} \equiv C\sum_{g\in\mathcal{V}_S} g$$

with a specific normalization constant C. In the Kitaev model, the plaquette operators $\{W_p\}$ serve as the generators of a stabilizer space $\langle\{W_p\}\rangle$, and the projector is exactly the projector preceding Eq. (1.1)

$$\hat{\Pi} = \left(\frac{\mathbb{1}+\tilde{W}_v}{2}\right)\left(\frac{\mathbb{1}+\tilde{W}_h}{2}\right)\prod_p\left(\frac{\mathbb{1}+W_p}{2}\right).$$

In other words, *the stabilizer space generated by the plaquette operators and the global Wilson loop operators is the flux-free subspace in the topological sector labeled by the number pair* $(\tilde{W}_v, \tilde{W}_h)$. Therefore, the dimension of such a stabilizer space is

$$\dim(\mathcal{V}_S) = \mathrm{Tr}(\hat{\Pi}) = 2^{N-N_p-1}.$$

Although the stabilizer space is not strictly one-dimensional, the model has only one ground state independent of the initial state. This implies that the matter part in Eq. (1.1) reduces the degrees of freedom through a mechanism different from the stabilizer projector. In the following discussion, we provide a physical picture where *the matter part is the non-stabilizerness of the ground state by examining the form of the stabilizer Rényi entropy.*

The *n-stabilizer Rényi entropy* is defined as [3],

$$M_n\left(|\psi\rangle\right) := \frac{1}{1-n}\ln\left(\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\frac{1}{2^N}\mathrm{Tr}\left(\rho\mathsf{S}\right)^{2n}\right) \tag{1.2}$$

where $\rho = |\psi\rangle\langle\psi|$ is the pure state density matrix of $|\psi\rangle$. We need to bear in mind that the definition above given in Ref. [3] applies strictly to pure states. For mixed states, such a definition will not be a valid monotone. For the Kitaev model, the stabilizer Rényi entropy can be calculated straightforwardly as follows

$$M_n\left(|\psi\rangle\right) = \frac{1}{1-n}\ln\left(\frac{1}{2^N}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\langle\mathsf{S}\rangle^{2n}\right) = \frac{1}{1-n}\ln\left(\frac{1}{2^{N+1}}\sum_{\mathfrak{D}}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n}\right) \tag{1.3}$$

In the derivation, we need to be careful that $\Sigma_{\mathfrak{D}_{zz}} \propto \Sigma_{\mathfrak{D}_Z}$. Owing to this, we have to divide the summation by two to avoid overcounting. Here we demonstrate the proof of the property that $\langle\mathsf{S}\rangle^2 = \eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2$ if $\mathsf{S} \propto \Sigma_{\mathfrak{D}}$, which has been used in the above derivation. Suppose $\Sigma_{\mathfrak{D}} = c\mathsf{S}$ with c a specific constant. By definition, the phase $\eta_{\mathfrak{D}}$ is given by $\eta_{\mathfrak{D}} = \langle\Sigma_{\mathfrak{D}}^2\rangle$. Substituting the assumption into the definition of $\eta_{\mathfrak{D}}$ gives

$$\langle\Sigma_{\mathfrak{D}}^2\rangle = \eta_{\mathfrak{D}} = \langle c^2\mathsf{S}^2\rangle = c^2$$

With this result, we ultimately have the following.

$$\langle\mathsf{S}\rangle^2 = \frac{1}{c^2}\langle\Sigma_{\mathfrak{D}}\rangle^2 = \eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2$$

This relation proves Eq. (1.3). The expectation value of the Pauli strings can be obtained by evaluating the Pfaffian of the correlation matrix defined by Eq. (C.16)

$$\langle\mathsf{S}\rangle^2 = \mathrm{Pf}\left(\mathsf{C}_{\mathfrak{D}}\right)^2.$$

Next, we exploit the organization of the density matrix to simplify the form of the SRE. Since both $\eta_{\mathfrak{D}}$ and $\mathfrak{C}_{\mathfrak{D}}$ depend only on equivalent classes of the strings, we have

$$\sum_{\mathfrak{D}}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n} = 2^{N_p+1}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n}$$

The factor 2^{N_p+1} comes from the number of flux sectors.

$$\frac{\dim(\mathcal{H})}{\dim(\hat{\Pi})} = \frac{2^N}{2^{N-N_p-1}} = 2^{N_p+1}$$

Substituting this result into the SRE formula, we obtain

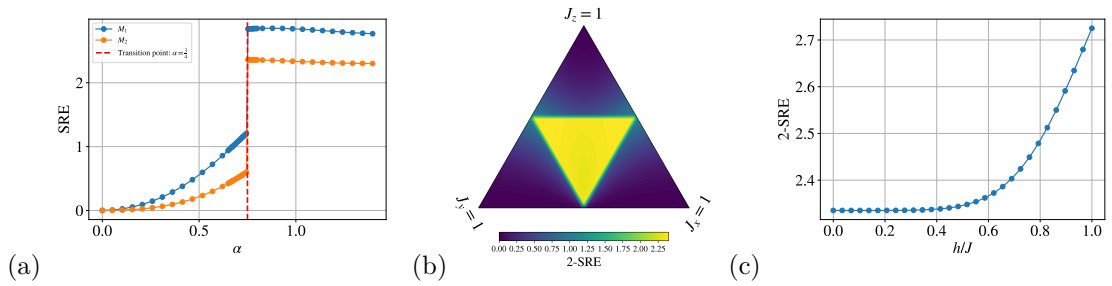
$$M_n\left(|\mathrm{Gnd}\rangle\right) = \frac{1}{1-n}\ln\left(\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n}\right) \tag{1.4}$$

As one would expect, the $n = 1$ stabilizer Rényi entropy is an indeterminate form due to the normalization condition Eq. (D.3). Therefore, computing the $n = 1$ entropy requires taking the limit.

$$M_1(|\mathrm{Gnd}\rangle) = \lim_{n\to1}\frac{1}{1-n}\ln\left(\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}^n\mathfrak{C}_{\mathfrak{D}}^{2n}\right) = -\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\ln\left(\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\right)$$

In the toric code limit, all non-zero coefficients $\mathfrak{C}_{\mathfrak{D}} = 1$ reduce to unity, **rendering** $M_n = 0$ [1]. This implies that the ground state is a perfect stabilizer state in the toric code limit, in agreement

[1] Note that there are $2^{N_d} = 2^{N/2} = 2^{N-N_p}$ configurations $[\mathfrak{D}]$ that give finite correlation functions $\mathfrak{C}_{\mathfrak{D}}$ in the toric code limit.



(a) (b) (c)

Figure 1: Numerical calculation of the stabilizer Rényi entropy (SRE) for the ground state of a 4×4 system. (a) The trend of SRE with α, which parameterizes the spin coupling by $J_x = J_y = \alpha/3$ and $J_z = 1 - 2\alpha/3$. The SRE jumps significantly at the gapped-gapless transition point. (b) The 2-SRE plotted over the parameter space as a phase diagram, showing a sharp boundary at the phase transition line. (c) The 2-SRE as a function of the external magnetic field h at the isotropic point $\alpha = 1$. However, we have to bear in mind that the result for the cases with an external magnetic field is valid only when h is sufficiently small so that perturbation theory is reasonable.

with our knowledge of the toric code ground state and its role as a stabilizer state. Eq. (1.3) shows that the stabilizer Rényi entropy depends only on the equivalence class of the edge configurations and the Majorana fermionic correlation functions, implying that the matter sector governs the magic resource of the ground state. Therefore, it can be sampled using fermionic techniques [4]. Fig. 1 shows numerical results for the stabilizer Rényi entropy.

2 Gauge-Matter Separation

Define the effective Majorana fermion density matrix

$$\varrho^F = \frac{1}{2^{N/2}} \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \mathcal{M}_{\mathfrak{D}}$$

$$M_n^F = \frac{1}{1-n} \ln \left(\sum_{[\mathfrak{D}]} \frac{1}{2^{N/2}} \mathrm{Tr}_{\mathcal{V}_S}^{2n} \left(\varrho^F c_{\mathfrak{D}} \right) \right)$$

$$\prod_{\langle ij \rangle} (-i c_i c_j) = (-i)$$

3 1-Form Wilson Symmetry, Gauss Law, and Magic Separation

4 Stabilizer Rényi Entropy of Reduced Density Matrices

The stabilizer Rényi entropy extends to general mixed states [3].

$$\tilde{M}_2(\rho) = -\ln \left(\frac{1}{2^N} \sum_{\mathsf{S} \in \tilde{\mathcal{P}}_N} \mathrm{Tr}\left(\rho \mathsf{S}\right)^4 \right) - R_2(\rho) \tag{4.1}$$

where $R_2(\rho) \equiv -\ln(\mathrm{Tr}\rho^2)$ is the 2-Rényi entropy of the state ρ. For the Kitaev model, the reduced density matrix of a subregion A reads [2]

$$\rho_A = \frac{1}{2^{N^A - N_p^A}} \hat{\Pi}_A \sum_{[\mathfrak{D}] \subset A} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}. \tag{4.2}$$

The 2-Rényi entropy can be calculated straightforwardly,

$$\begin{aligned}
\mathrm{Tr}_A\rho_A^2 &= \frac{1}{2^{2\left(N^A-N_p^A\right)}}\sum_{[\mathfrak{D}]\subset A}\sum_{[\mathfrak{D}']\subset A}\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}'}\mathrm{Tr}_A\left(\hat{\Pi}_A\Sigma_{\mathfrak{D}}\Sigma_{\mathfrak{D}'}\right)\\
&= \frac{1}{2^{2\left(N^A-N_p^A\right)}}\sum_{[\mathfrak{D}]\subset A}\sum_{[\mathfrak{D}']\subset A}\vartheta\left(\mathfrak{D},\mathfrak{D}'\right)\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}'}\mathrm{Tr}_A\left(\hat{\Pi}_A\Sigma_{\mathfrak{D}\ominus\mathfrak{D}'}\right)\\
&= \frac{1}{2^{2\left(N^A-N_p^A\right)}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\mathrm{Tr}_A\hat{\Pi}_A\\
&= \frac{1}{2^{N^A-N_p^A}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2.
\end{aligned}$$

The first term of Eq. (4.1) can also be expressed in terms of the coefficients $\mathfrak{C}_{\mathfrak{D}}$.

$$\frac{1}{2^{N^A}}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_{N^A}}\left[\mathrm{Tr}_A\left(\rho_A\mathsf{S}\right)\right]^4 = \frac{1}{2^{N^A}}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_{N^A}}\langle\mathsf{S}\rangle^4 = \frac{1}{2^{N^A}}\sum_{\mathfrak{D}\subset A}\eta_{\mathfrak{D}}^2\mathfrak{C}_{\mathfrak{D}}^4 = \frac{1}{2^{N^A-N_p^A}}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4$$

Together, the SRE for the reduced density matrix reduces to the following form,

$$\tilde{M}_2\left(\rho_A\right) = -\ln\left(\frac{\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4}{\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2}\right) = -\ln\left(\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4\right) + \ln\det\left(\mathbb{1}+\mathsf{C}_A\right). \tag{4.3}$$

This form has already been proposed in Ref. [4].

Now, let's turn to the Pauli spectrum.

Question: Does $\tilde{M}(\rho_A)=0$ implies $M(|\psi\rangle)=0$?

Feasible Outlook

- Clarify the layers of *organization*. In this way, we can discuss quantum magic and entanglement on equal footing [**?**].
- To examine whether the block-diagonal structure of the RDM implies deeper physics about quantum magic reflected by calculating the SRE.
- What would happen to SRE if a pair of visons is created? Could we see information about anyonic statistics from the resulting SRE [**?**]?

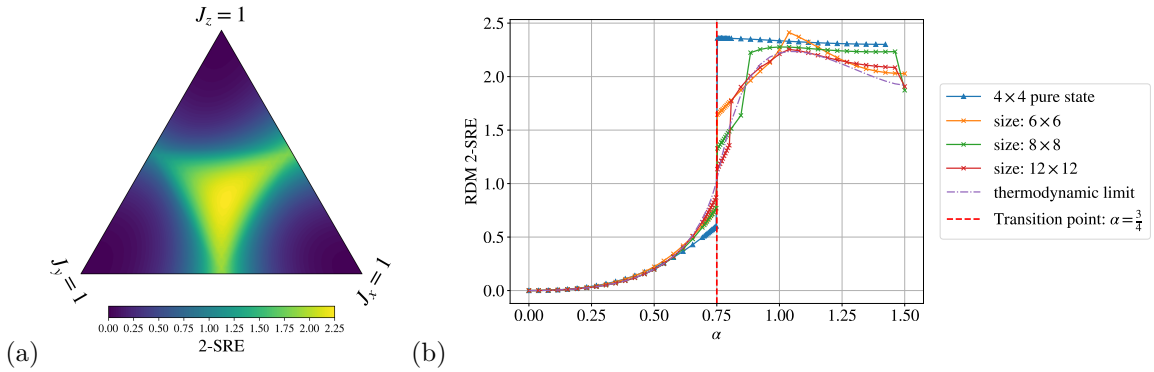


Figure 2: The SRE of a subregion illustrated in Fig. 4.c. (a) The ternary phase diagram mapping the 2-SRE across the coupling parameter space. (b) Comparison of pure-state SRE and mixed-state SRE of the region with size 4×4. In finite-size systems, the SRE as a function of coupling parameters shows a discontinuity at the gapped-gapless transition point. However, the jump decreases as the system size increases and eventually closes in the thermodynamic limit.

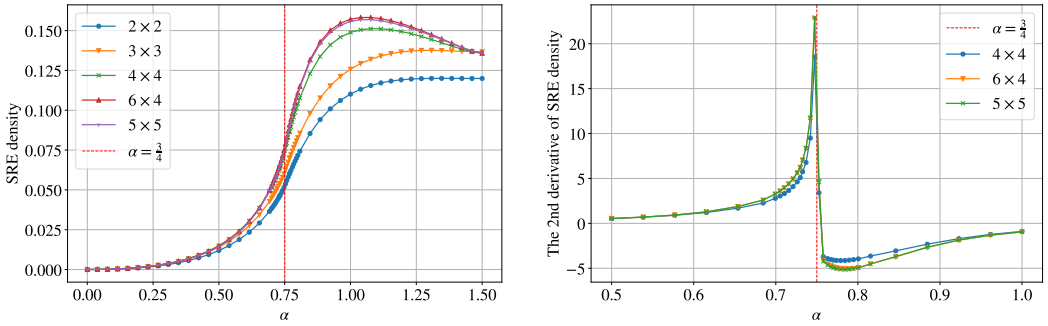


Figure 3: The subsystem SRE density in the thermodynamic limit as a function of α with different subsystem sizes.

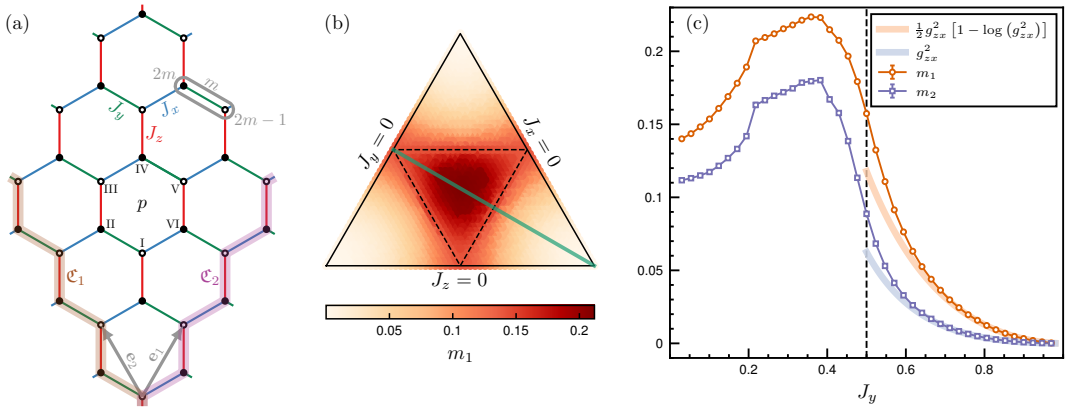


Figure 4: Taken from Ref. [**?**]

5 Magic of Non-Simply Connected Subsystems

The reduced density matrix of a subsystem with a hole inside takes the following form

$$\rho_A = \frac{1}{2^{N^A - N_p^A - 2g}} \hat{\Pi}_A \hat{\Pi}_{\mathcal{L}} \sum_{[\mathfrak{D}] \subset A} \left(\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}} \Sigma_{\mathfrak{D} \ominus \{\mathfrak{H}\}} \right).$$

The summation of the expectation value of Pauli strings to the power four is then

$$\frac{1}{2^{N^A}} \sum_{\mathsf{S} \in \tilde{\mathcal{P}}_{N^A}} \left[\mathrm{Tr}_A \left(\rho_A \mathsf{S} \right) \right]^4 = \frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \left[\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \left(\frac{\mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}}{\mathfrak{C}_{\mathfrak{D}}} \right)^4 \right] \mathfrak{C}_{\mathfrak{D}}^4$$

Because $\mathfrak{C}_{\mathfrak{D}}^2 \geq \mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}^2$, the series in the square bracket is less than unity,

$$\frac{1}{2^g} \sum_{\{\mathfrak{H}\}} \left(\frac{\mathfrak{C}_{\mathfrak{D} \ominus \{\mathfrak{H}\}}}{\mathfrak{C}_{\mathfrak{D}}} \right)^4 \leq \frac{1}{2^g} \sum_{n=1}^{2^g} = 1.$$

Therefore, the magic part of the subsystem 2-SRE is greater than that computed by the Majorana state with the topological boundary correction,

$$- \ln \left[\frac{1}{2^{N^A}} \sum_{\mathsf{S} \in \tilde{\mathcal{P}}_{N^A}} \left[\mathrm{Tr}_A \left(\rho_A \mathsf{S} \right) \right]^4 \right] \geq - \ln \left(\frac{1}{2^{N^A - N_p^A - 2g}} \sum_{[\mathfrak{D}] \subset A} \mathfrak{C}_{\mathfrak{D}}^4 \right).$$

6 Stabilizer States, Polytope and Magic

A stabilizer state of an n-qubit system is determined by the generators $\{P_1, P_2, \cdots, P_n\}$ of the stabilizer group $\mathcal{G} = \langle P_1, P_2, \cdots, P_n \rangle$ in the following way,

$$\zeta_{\mathcal{G}} = \prod_{P \in \mathcal{G}} \left(\frac{\mathbb{1} + P}{2} \right)$$

In other words, a stabilizer state uniquely corresponds to a set of stabilizer generators.

$$\mathrm{STAB} = \left\{ \sum_{\mathcal{G}} x_{\mathcal{G}} \zeta_{\mathcal{G}} \;\middle|\; x_{\mathcal{G}} \geq 0, \; \sum_{\mathcal{G}} x_{\mathcal{G}} = 1 \right\}$$

This set is also called the *stabilizer polytope*, and **a state lying in the stabilizer polytope is magic-free**. We would like to point out that every point in the set STAB is a valid density matrix. This is because the stabilizer states $\zeta_{\mathcal{G}}$ that generate STAB are valid pure-state density matrices, and an arbitrary convex mixture of density matrices is also a valid density matrix. On the other hand, this definition also broadens the original definition of stabilizer states to mixed states.

$$\mathcal{R}[\rho] = \min_{\{x_l\}} \left\{ \sum_l |x_l| \;\middle|\; \rho = \sum_{\mathcal{G}} x_{\mathcal{G}} \zeta_{\mathcal{G}}, \; x_{\mathcal{G}} \in \mathbb{R}, \; \sum_{\mathcal{G}} x_{\mathcal{G}} = 1 \right\} \tag{6.1}$$

Note that the quantity interprets the negativity since

$$\sum_{\mathcal{G}} |x_{\mathcal{G}}| = \sum_{x_{\mathcal{G}}>0} x_{\mathcal{G}} - \sum_{x_{\mathcal{G}}\leq 0} x_{\mathcal{G}} = 1 + 2\sum_{x_{\mathcal{G}}\leq 0} |x_{\mathcal{G}}|$$

Because a magical state cannot be represented as a convex mixture of stabilizer states, $\{x_{\mathcal{G}}\}$ must have negative components for a magical state ρ to be represented in the form given in (**??**). Therefore, the RoM is larger than one $\mathcal{R}[\rho] > 1$ if the state ρ is magical, making it a measure of non-stabilizerness.

$$b_j \equiv \mathrm{Tr}\left(\rho \mathsf{S}_j\right) = \sum_{\mathcal{G}} x_{\mathcal{G}} \mathrm{Tr}\left(\zeta_{\mathcal{G}} \mathsf{S}_j\right) \equiv \sum_{\mathcal{G}} A_{j\mathcal{G}} x_{\mathcal{G}}$$

$$\sum_j A_{\mathcal{G}j} A_{j\mathcal{G}} = \sum_j \langle \zeta_{\mathcal{G}} | \mathsf{S}_j | \zeta_{\mathcal{G}} \rangle^2 = 2^n \Longrightarrow A^T A = 2^n \mathbf{1}_n$$

$$A^T \mathbf{b} = \mathbf{x}$$

$$\mathcal{R}[\rho] = \min_{\mathbf{z}} \left\{ \mathbf{1}^T \mathbf{z} \mid -\mathbf{z} \leq \mathbf{x} \leq \mathbf{z},\ \mathbf{x} = A^{-1}\mathbf{b} \right\}$$

$$\mathcal{R}[\rho] = \max_{\mathbf{y}} \left\{ \mathbf{b}^T \mathbf{y} \mid -\mathbf{1} \leq A^T \mathbf{y} \leq \mathbf{1} \right\}$$

Define

$$W_{\mathbf{y}} = \sum_j y_j \mathsf{S}_j. \tag{6.2}$$

The dual condition implies

$$|\langle \zeta_{\mathcal{G}} | W | \zeta_{\mathcal{G}} \rangle| = |\mathrm{Tr}\left(W \zeta_{\mathcal{G}}\right)| = \left| \sum_j y_j \mathrm{Tr}\left(\zeta_{\mathcal{G}} \mathsf{S}_j\right) \right| = \left| A^T \mathbf{y} \right| \leq 1\ , \quad \forall \mathcal{G}$$

At the optimal point of this linear program, we obtain the optimal magic witness

$$W^* = \sum_j y_j^* \mathsf{S}_j,$$

which satisfies

$$\mathcal{R}[\rho] = \mathrm{Tr}\left(W^* \rho\right) = \sum_j y_j^* \mathrm{Tr}\left(\rho \mathsf{S}_j\right) = \mathbf{b}^T \mathbf{y}^*$$

The paper [**?**] investigates the structure of the optimal witness operator W^* at the onset temperature where quantum magic starts to take shape. It happens that the magic onset temperature is near the critical temperature of the phase transition between the paramagnet and the phase where the Majorana fermions are itinerant.

A Minimal Size for Non-trivial Topology

For numerical efficiency, we tend to choose the smallest possible system. However, the system must still be sufficiently large such that the topological order still exists. A requirement for this is

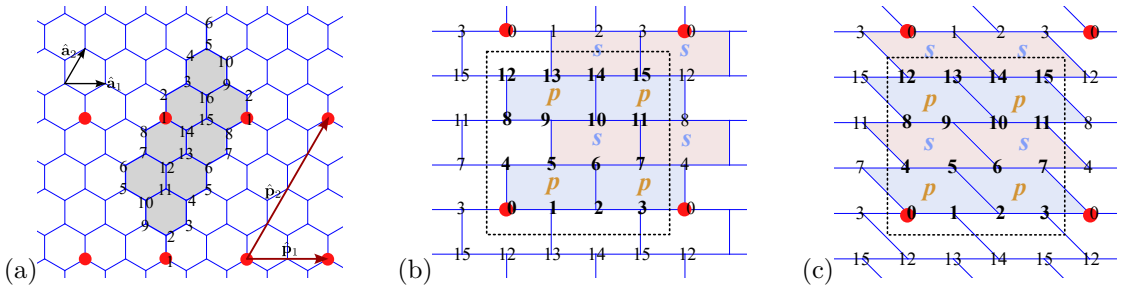


Figure 5: (a) The smallest topologically non-trivial system consists of 8 unit cells, which is equivalent to 16 spins. The two vectors that define the periodic boundary condition are $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, respectively. (b) For numerical simplicity, the honeycomb lattice is rearranged into a brick-wall geometry, and the spins are relabeled according to their Euclidean coordinates in the lattice. In both figures, red dots indicate identical lattice sites. (c) An equivalent choice.

that the ground state manifold should possess four independent topological sectors generated by global Wilson loops if the geometry of the system is a torus.

We start from the Wen plaquette model. There are two requirements for a Wen plaquette system to possess four topologically degenerate ground states:

- The lattice must contain an even number of rows or columns in both the horizontal and vertical directions.
- A global Wilson loop cannot be generated or destroyed by local plaquette operators.

The physical meaning of the first requirement is to preserve the checkerboard pattern of the lattice. If there are adjacent plaquettes (cells) that are of the same type (black or white, in a checkerboard sense), then the two types of plaquettes become identical, and the original two different Wilson loops become the same. This breakdown of the checkerboard pattern reduces the original fourfold topological ground state degeneracy down to twofold, which is not adequate to describe the physics of a toric code system.

According to the two requirements, **the smallest possible topologically non-trivial Wen plaquette model contains eight** ($8 = 2 \times 4$) **distinct plaquettes and eight sites**. A 2×2 checkerboard is not adequate for the system to have four topological sectors since it violates the second requirement[2].

B 1D Kitaev Model

Jordan-Wigner Transform

$$\sigma_i^z = i\bar{c}_i c_i, \quad \sigma_i^x = \left(\prod_{j<i} i\bar{c}_j c_j\right) c_i, \quad \sigma_i^y = \left(\prod_{j<i} i\bar{c}_j c_j\right) \bar{c}_i$$

$$H = \sum_r J_x \sigma_{rA}^x \sigma_{rB}^x + J_y \sigma_{rB}^y \sigma_{r+1,A}^y = \sum_r \left(-iJ_x \bar{c}_{rA} c_{rB} + iJ_y c_{rB} \bar{c}_{r+1,A}\right)$$

$$\begin{cases} \sigma_{rA}^x \sigma_{rB}^x = -i\bar{c}_{rA} c_{rB} \\ \sigma_{rB}^y \sigma_{r+1,A}^y = i c_{rB} \bar{c}_{r+1,A} \end{cases}$$

[2]The *global Wilson loop* is identical to a plaquette operator

$$\left(\sigma_{rA}^{x}\sigma_{rB}^{x}\right)\left(\sigma_{rB}^{y}\sigma_{r+1,A}^{y}\right)=\bar{c}_{rA}\bar{c}_{r+1,A}$$

Valid correlators should be of the form $\langle\bar{c}_{iA}c_{jB}\rangle$.

$$\begin{cases}\sigma_{rA}^{x}\sigma_{rB}^{y}=-i\bar{c}_{rA}\bar{c}_{rB}\\ \sigma_{rB}^{x}\sigma_{r+1,A}^{y}=i\bar{c}_{rB}\bar{c}_{r+1,A}\end{cases}$$

Note that the Wilson loop operator is equivalent to the total fermion parity, which is a conserved quantum number.

$$\begin{aligned}\prod_{\langle ij\rangle}\sigma_{i}^{\alpha_{ij}}\sigma_{j}^{\alpha_{ij}}&=\left(\prod_{r=1}^{N-1}\sigma_{rA}^{x}\sigma_{rB}^{x}\sigma_{rB}^{y}\sigma_{r+1,A}^{y}\right)\sigma_{NA}^{x}\sigma_{NB}^{x}\sigma_{NB}^{y}\sigma_{1A}^{y}\\&=\bar{c}_{1A}\bar{c}_{NA}\left(-i\bar{c}_{NA}c_{NB}\right)\left(\prod_{j<NB}i\bar{c}_{j}c_{j}\right)\bar{c}_{NB}\bar{c}_{1A}\\&=-\prod_{j}i\bar{c}_{j}c_{j}=-\prod_{j}\sigma_{j}^{z}=-(-1)^{F}\equiv-\tilde{Z}\end{aligned}$$

We have to choose one of the sectors to fix the fermion parity. This is equivalent to fixing the superselection sector. In addition, the model possesses a $\mathbb{Z}_2$ symmetry, so the following operator commutes with the Hamiltonian and serves as a global symmetry operation

$$\tilde{X}\equiv\prod_{i}\sigma_{i}^{x}=\prod_{r}\left(-i\bar{c}_{rA}c_{rB}\right).$$

Therefore, we can also fix the eigenvalue of $\tilde{X}$ to be either $+1$ or -1. As a result, **the ground state is four-fold degenerate, and one needs to select one of the superselection sectors labeled by** $\left\{\tilde{X},\tilde{Z}\right\}$.

$$H=\sum_{r}\begin{pmatrix}\bar{c}_{kA}\\ c_{kB}\end{pmatrix}^{\dagger}\begin{pmatrix}0 & s_{k}\\ s_{k}^{*} & 0\end{pmatrix}\begin{pmatrix}\bar{c}_{kA}\\ c_{kB}\end{pmatrix}$$

where $s_k=-iJ_x+iJ_ye^{-ik}$.

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & e^{i\theta_k}\\ e^{-i\theta_k} & -1\end{pmatrix}\begin{pmatrix}0 & s_{k}\\ s_{k}^{*} & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}1 & e^{i\theta_k}\\ e^{-i\theta_k} & -1\end{pmatrix}=\begin{pmatrix}+|s_k| & 0\\ 0 & -|s_k|\end{pmatrix}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix}1 & e^{i\theta_k}\\ e^{-i\theta_k} & -1\end{pmatrix}\begin{pmatrix}\bar{c}_{kA}\\ c_{kB}\end{pmatrix}=\frac{1}{\sqrt{2}}\begin{pmatrix}\bar{c}_{kA}+e^{i\theta_k}c_{kB}\\ e^{-i\theta_k}\bar{c}_{kA}-c_{kB}\end{pmatrix}\equiv\begin{pmatrix}b_k\\ a_k\end{pmatrix}$$

The ground state of the model is the state with the lower band a_k fully occupied. Therefore, the correlation function of a_k satisfies $\langle a_k^{\dagger}a_{k'}\rangle=\delta_{kk'}$ and $\langle a_k a_{k'}\rangle=0$. With this, the correlation

function of the Majorana fermions is

$$\begin{aligned}\langle \bar{c}_{iA} c_{jB}\rangle &= \frac{1}{N}\sum_{k,k'} e^{i(kj-k'i)}\langle \bar{c}^{\dagger}_{kA} c_{k'B}\rangle \\ &= \frac{1}{N}\sum_{k,k'} e^{i(kj-k'i)}\frac{1}{2}\langle\left(b^{\dagger}_{k}+e^{-i\theta_k}a^{\dagger}_{k}\right)\left(e^{-i\theta_k}b_{k'}-a_{k'}\right)\rangle \\ &= -\frac{1}{2N}\sum_{k} e^{i[k(j-i)-\theta_k]}\end{aligned}$$

Note that N denotes the number of Bravais lattices, which is equal to half of the number of lattice sites in this case.

$$\langle i\bar{c}_{iA} c_{jB}\rangle = \frac{1}{2N}\sum_{k}\frac{J_x\cos\left[k\left(j-i\right)\right]-J_y\cos\left[k\left(j-i+1\right)\right]}{\sqrt{J_x^2+J_y^2-2J_xJ_y\cos k}} \tag{B.1}$$

In the continuum limit, the correlation function can be obtained iteratively. The function I_m defined as

$$I_m \equiv \frac{1}{2\pi}\int_{-\pi}^{\pi}\frac{\cos(mk)}{\sqrt{J_x^2+J_y^2-2J_xJ_y\cos k}}\mathrm{d}k$$

satisfies the recurrence relation

$$(2m+1)J_xJ_yI_{m+1}-2m(J_x^2+J_y^2)I_m+(2m-1)J_xJ_yI_{m-1}=0$$

In the 1D model, the fermions $\bar{c}_{rB}$ and c_{rA} are not included in the Hamiltonian, so they are redundant degrees of freedom.

$$\begin{cases}\sigma^x_{rB}\sigma^x_{r+1,A} = -i\bar{c}_{rB}c_{r+1,A}\\ \sigma^y_{rA}\sigma^y_{rB} = ic_{rA}\bar{c}_{rB}\end{cases}$$

Note that the ξ operators also commute with the Hamiltonian. Therefore, the ground state is actually extensively degenerate, and we need to apply the projector to pin down the sectors. Otherwise, the ground state will be a mixed state.

$$\rho = C\left(\frac{\mathbb{1}+\tilde{Z}}{2}\right)\prod_{\hat{\xi}\in\Omega}\left(\frac{\mathbb{1}+\hat{\xi}}{2}\right)\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}}$$

The $\hat{\xi}$ operators are defined as follows.

$$\hat{\xi}_{ij} = \sigma^z_i\Sigma_{\overline{ij}}\sigma^z_j$$

We have to note that different choices of the spatial arrangement of the $\hat{\xi}$ operators correspond to different ways of dividing the Hilbert space into degenerate topological sectors for the 1D system. Consider two different ways, Ω and Ω', of defining $\hat{\xi}$ operators, where Ω set contains $\hat{\xi}_{\overline{rA,rB}}$ and Ω' contains $\hat{\xi}_{\overline{rB,(r+1)A}}$. The Pauli algebra implies that the two operators anticommute: $\left\{\hat{\xi}_{\overline{rA,rB}},\hat{\xi}_{\overline{rB,(r+1)A}}\right\}=0$. Therefore, the operator $\hat{\xi}_{\overline{rB,(r+1)A}}$ in Ω' can flip the operator $\hat{\xi}_{\overline{rA,rB}}$ in Ω, which is equivalent to transform the state labeled by the xi operators arranged in Ω with $\xi_{\overline{rA,rB}}$ in $\Omega=+1$ to the one with $\xi_{\overline{rA,rB}}$ in $\Omega=+1$. This pattern is similar to the toric code model, where

we also have multiple choices to define the four degenerate topological sectors. We emphasize that the choice of the spatial arrangement Ω for the $\hat{\xi}$ operators is not merely a gauge convention, but a manual selection of the topological sector within the degenerate ground-state manifold. Just as in the toric code, where choosing different Wilson loop invariants (W_x vs. W_z) selects distinct physical ground states with different entanglement spectra, defining $\hat{\xi}$ via Ω versus Ω' fixes a specific state in the 1D topological sector. Because these sector-labeled states are non-trivial superpositions of each other, their entangling properties across a spatial boundary ∂A naturally reflect the chosen topological sector.

We also have to point out that string operators and the ξ operators are not completely independent, since the standard arrangement of defining the ξ operators leads to the following identity

$$\prod_i \hat{\xi}_i \propto \tilde{X} \propto \Sigma_{\mathfrak{D}_{\tilde{X}}} \tag{B.2}$$

$$\mathrm{Tr}\rho = \frac{C}{2^{N_\xi+1}} \left(\mathfrak{C}_\varnothing + \mathfrak{C}_{\mathfrak{D}_{\tilde{X}}}\right) 2^N = 1$$

Therefore, the density matrix is

$$\rho = \frac{1}{2^{N-N_\xi}} \left(\frac{\mathbb{1}+\tilde{Z}}{2}\right) \prod_{\hat{\xi}\in\Omega} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}} \tag{B.3}$$

$$\begin{aligned}
\rho^2 &= \frac{1}{2^{2(N-N_\xi)}} \left(\frac{\mathbb{1}+\tilde{Z}}{2}\right) \prod_{\hat{\xi}\in\Omega} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}],[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}'} \varphi(\mathfrak{D},\mathfrak{D}') \Sigma_{\mathfrak{D}\ominus\mathfrak{D}'} \\
&= \frac{1}{2^{2(N-N_\xi)}} \left(\frac{\mathbb{1}+\tilde{Z}}{2}\right) \prod_{\hat{\xi}\in\Omega} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}''],[\mathfrak{D}']} \mathfrak{C}_{\mathfrak{D}''\ominus\mathfrak{D}'} \mathfrak{C}_{\mathfrak{D}'} \varphi(\mathfrak{D}''\ominus\mathfrak{D}',\mathfrak{D}') \Sigma_{\mathfrak{D}''} \\
&= \rho
\end{aligned}$$

$$\sum_{[\mathfrak{D}]} \eta_{\mathfrak{D}} \mathfrak{C}_{\mathfrak{D}}^2 = 2^{N-N_\xi}$$

The reduced density matrix can be constructed straightforwardly

$$\begin{aligned}
\rho_A = \mathrm{Tr}_{\overline{A}}\rho &= \frac{1}{2^{N-N_\xi^A+1}} \prod_{\hat{\xi}\in\tilde{\Omega}_A} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \mathrm{Tr}_{\overline{A}} \left[\prod_{\hat{\xi}\in\Omega} \left(\mathbb{1}+\hat{\xi}\right) \Sigma_{\mathfrak{D}}\right] \\
&= \frac{1}{2^{N-N_\xi^A+1}} \prod_{\hat{\xi}\in\tilde{\Omega}_A} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}} \mathrm{Tr}_{\overline{A}} \left[\left(\mathbb{1}+\tilde{X}\right) \Sigma_{\mathfrak{D}}\right] \\
&= \frac{1}{2^{N^A-N_\xi^A}} \prod_{\hat{\xi}\in\tilde{\Omega}_A} \left(\frac{\mathbb{1}+\hat{\xi}}{2}\right) \sum_{[\mathfrak{D}]\subset A} \mathfrak{C}_{\mathfrak{D}} \Sigma_{\mathfrak{D}}
\end{aligned}$$

But the choice and the pattern of the arrangement of the ξ operators are a kind of artificial.

$$\begin{aligned}\mathrm{Tr}_A\rho_A^2 &= \frac{1}{2^{2\left(N^A-N_\xi^A\right)}}\sum_{[\mathfrak{D}],[\mathfrak{D}']\subset A}\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{D}'}\varphi(\mathfrak{D},\mathfrak{D}\ominus\mathfrak{D}')\mathrm{Tr}_A\left[\prod_{\hat{\xi}\in\tilde{\Omega}_A}\left(\frac{\mathbb{1}+\hat{\xi}}{2}\right)\Sigma_{\mathfrak{D}'}\right]\\ &= \frac{1}{2^{N^A-N_\xi^A}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\end{aligned}$$

Therefore, the 2-Rényi entropy reads

$$R_2(\rho_A) = -\ln\left(\frac{1}{2^{N^A-N_\xi^A}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\right) \tag{B.4}$$

The gauge contribution to the entanglement entropy is zero.

$$\left(\frac{1}{2}\left|\partial A\right|-1\right) = \frac{1}{2}\times 2 - 1 = 0$$

Now we turn to the stabilizer Rényi entropy. The pure-state SRE can be obtained straightforwardly. The expectation value of each Pauli string is

$$\begin{aligned}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\mathrm{Tr}^4(\rho\mathsf{S}) &= \sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\mathrm{Tr}^4(\rho\mathsf{S})\\ &= \frac{1}{2^{4N+4}}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_N}\left\{\sum_{\varpi\subset\Omega}\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}\mathrm{Tr}\left[\left(\mathbb{1}+\tilde{Z}\right)\left(\prod_{\hat{\xi}\in\varpi}\hat{\xi}\right)\Sigma_{\mathfrak{D}}\mathsf{S}\right]\right\}^4\\ &= \frac{1}{2^4}2^{N_\xi+2}\sum_{[\mathfrak{D}]}\mathfrak{C}_{\mathfrak{D}}^4\end{aligned}$$

$$\begin{aligned}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_{N^A}}\mathrm{Tr}_A^4(\rho_A\mathsf{S}) &= \frac{1}{2^{4N^A}}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_{N^A}}\left\{\sum_{\varpi\subset\tilde{\Omega}_A}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}\mathrm{Tr}_A\left[\left(\prod_{\hat{\xi}\in\varpi}\hat{\xi}\right)\Sigma_{\mathfrak{D}}\mathsf{S}\right]\right\}^4\\ &= 2^{N_\xi^A}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4\end{aligned}$$

$$M_2(\rho_A) \equiv -\ln\left(\frac{1}{2^{N^A}}\sum_{\mathsf{S}\in\tilde{\mathcal{P}}_{N^A}}\mathrm{Tr}^4\left(\rho_A\mathsf{S}\right)\right) = -\ln\left(\frac{1}{2^{N^A-N_\xi^A}}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4\right) \tag{B.5}$$

$$\tilde{M}_2 = -\ln\left(\frac{1}{2^{N^A-N_\xi^A}}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4\right) + \ln\left(\frac{1}{2^{N^A-N_\xi^A}}\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2\right) = -\ln\left(\frac{\sum_{[\mathfrak{D}]\subset A}{\mathfrak{C}_{\mathfrak{D}}}^4}{\det\left(\mathbb{1}+\mathsf{C}_A\right)}\right) \tag{B.6}$$

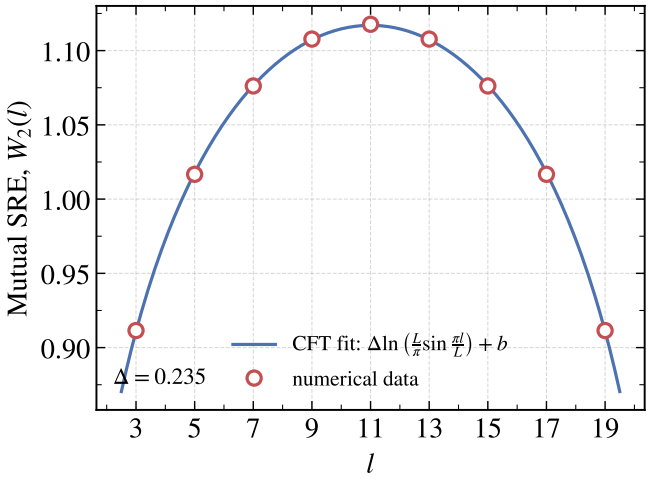


Figure 6: The mutual SRE of the 1D Kitaev model at the gapless point as a function of division. The equation satisfies $W_2(l) = \frac{1}{4}\ln l_c$.

C Finite-Size Correlation

As mentioned in our work [2], at the ground state, the multi-spin correlation function of the Kitaev model is finite if and only if it can be written in the form of a string. Therefore, we only need to discuss the following form of the correlation function

$$\left\langle \prod_{\langle ij\rangle\in\mathfrak{D}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \right\rangle$$

This kind of correlation function can be exactly evaluated in the Majorana representation. Substituting the transformation

$$\sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \to ib_i^{\alpha_{ij}}c_i ib_j^{\alpha_{ij}}c_j = \hat{u}_{ij}ic_jc_i$$

into the correlation function and taking the trivial gauge, we have

$$\left\langle \prod_{\langle ij\rangle\in\mathfrak{D}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \right\rangle = \left\langle \prod_{\langle ij\rangle\in\mathfrak{D}} (-i)c_ic_j \right\rangle_M$$

Therefore, we only need to focus on the two-point correlation functions. Here we set the system to satisfy the **periodic boundary condition**, which is characterized by two independent vectors, $\hat{\mathbf{p}}_1 = 2\hat{\mathbf{a}}_1$ and $\hat{\mathbf{p}}_2 = 4\hat{\mathbf{a}}_2$, as illustrated in Fig.4.a. The geometry adopted here follows the one illustrated in Ref. [5]

C.1 Momentum States in Finite Systems

For slave fermions in the parton construction, the system boundary condition can be periodic (PBC) or anti-periodic (APBC), depending on the chosen topological sector [6]. Here, we only consider the PBC in both horizontal and vertical directions.

$$f_{\sigma\mu\mathbf{x}} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_1)} = f_{\sigma\mu,(\mathbf{x}+\hat{\mathbf{p}}_2)} \tag{C.1}$$

To determine the allowed momentum states, we first Fourier transform the fermions as follows

$$f_{\sigma\mu\mathbf{x}} = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} f_{\sigma\mu\mathbf{k}}$$

With this expression, Eq. (C.1) implies that the allowed momentum $\mathbf{k}$ must satisfy

$$\begin{cases} \mathbf{k}\cdot\hat{\mathbf{p}}_1 = 2n_1\pi \quad ; \quad n_1 = 0, 1, \cdots, N_h - 1 \\ \mathbf{k}\cdot\hat{\mathbf{p}}_2 = 2n_2\pi \quad ; \quad n_2 = 0, 1, \cdots, N_v - 1 \end{cases} \tag{C.2}$$

In our case illustrated in Fig. 4, $N_h = 2$ and $N_v = 4$, representing the number of unit cells included in the system in the horizontal and vertical directions, respectively. Consider the unit vectors $\hat{\mathbf{a}}_1 = \hat{\mathbf{e}}_x$ and $\hat{\mathbf{a}}_2 = \frac{1}{2}\hat{\mathbf{e}}_x + \frac{\sqrt{3}}{2}\hat{\mathbf{e}}_y$. We can define the unit vectors in the momentum space as

$$\begin{cases} \hat{\mathbf{b}}_1 = 2\pi\hat{\mathbf{e}}_x - \dfrac{2\pi}{\sqrt{3}}\hat{\mathbf{e}}_y \\ \hat{\mathbf{b}}_2 = \dfrac{4\pi}{\sqrt{3}}\hat{\mathbf{e}}_y \end{cases}$$

such that $\hat{\mathbf{a}}_i \cdot \hat{\mathbf{b}}_j = 2\pi\delta_{ij}$. With these basis vectors, the allowed momentum states are

$$\mathbf{k} = \frac{n_1}{2}\hat{\mathbf{b}}_1 + \frac{n_2}{4}\hat{\mathbf{b}}_2 \quad ; \quad n_1 = 0, 1 \quad , \quad n_2 = 0, 1, 2, 3 \tag{C.3}$$

In general, if the relation between two representations $\mathbf{k} = k_x\hat{\mathbf{e}}_x + k_y\hat{\mathbf{e}}_y = k_1\hat{\mathbf{b}}_1 + k_2\hat{\mathbf{b}}_2$ is

$$\begin{cases} k_x = 2\pi k_1 \\ k_y = \dfrac{2\pi}{\sqrt{3}}\left(-k_1 + 2k_2\right) \end{cases} , \quad \begin{cases} k_1 = \dfrac{1}{2\pi}k_x \\ k_2 = \dfrac{1}{4\pi}\left(k_x + \sqrt{3}k_y\right) \end{cases}$$

C.2 Two-Points Correlation Functions

The correlation functions can be calculated more easily in the normal fermion language. As defined in the appendix, the c-Majorana fermions can be written in terms of γ-fermions,

$$\begin{cases} c_{\mathbf{x}A} = \gamma^\dagger_{\mathbf{x}} + \gamma_{\mathbf{x}} = \dfrac{1}{\sqrt{N/2}} \displaystyle\sum_{\mathbf{k}} \left(\gamma^\dagger_k e^{-i\mathbf{k}\mathbf{x}} + \gamma_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}}\right) \\ c_{\mathbf{x}B} = i\left(\gamma^\dagger_{\mathbf{x}} - \gamma_{\mathbf{x}}\right) = \dfrac{i}{\sqrt{N/2}} \displaystyle\sum_{\mathbf{k}} \left(\gamma^\dagger_k e^{-i\mathbf{k}\mathbf{x}} - \gamma_{\mathbf{k}} e^{i\mathbf{k}\mathbf{x}}\right) \end{cases}$$

And the ground state is

$$|\psi\rangle = A \exp\left(\sum_{k\in \mathrm{HBZ}} g_k \gamma^\dagger_k \gamma^\dagger_{-k}\right)|0\rangle$$

where

$$A = \prod_{k\in \mathrm{HBZ}} \left(1 + |g_k|^2\right)^{-1/2}$$

is the normalization constant. Similar to Numbu-Gor'kov Green's function, there are two types of fermion pairing, $AA(BB)$ and AB. Both types of correlation functions can be simplified to

the problem of solving the γ-fermion correlation function as we will see later. The two γ-fermion correlation functions we have to know are $\langle\gamma_{\mathbf{k}}\gamma_{k'}\rangle$ and $\langle\gamma_k^\dagger\gamma_{k'}\rangle$. Their analytical forms are given as follows

$$\begin{cases}\langle\gamma_k^\dagger\gamma_{k'}\rangle=\dfrac{|g_k|^2}{1+|g_k|^2}\delta_{kk'}=v_k^2\delta_{kk'}=\dfrac{1}{2}\left(1-\dfrac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|}\right)\delta_{kk'}\\ \langle\gamma_{\mathbf{k}}\gamma_{k'}\rangle=\dfrac{g_k}{1+|g_k|^2}\delta_{k,-k'}=i\mathrm{sgn}\left(I_k\right)u_kv_k\delta_{k,-k'}=i\dfrac{I_k}{2\left|s_{\mathbf{k}}\right|}\delta_{k,-k'}\end{cases}$$

and the fermionic commutation relations can give other two types of correlation functions

$$\begin{cases}\langle\gamma_{\mathbf{k}}\gamma_{k'}^\dagger\rangle=\delta_{kk'}-\langle\gamma_{k'}^\dagger\gamma_{\mathbf{k}}\rangle=u_k^2\delta_{kk'}=\dfrac{1}{2}\left(1+\dfrac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|}\right)\delta_{kk'}\\ \langle\gamma_{\mathbf{k}}^\dagger\gamma_{k'}^\dagger\rangle=\langle\gamma_{k'}\gamma_{\mathbf{k}}\rangle^*=-i\mathrm{sgn}(I_{\mathbf{k}'})u_{\mathbf{k}'}v_{\mathbf{k}'}\delta_{\mathbf{k}',-\mathbf{k}}=i\mathrm{sgn}(I_k)u_{\mathbf{k}}v_{\mathbf{k}}\delta_{\mathbf{k},-\mathbf{k}'}=\langle\gamma_{\mathbf{k}}\gamma_{\mathbf{k}'}\rangle\delta_{\mathbf{k},-\mathbf{k}'}\end{cases}$$

Here I list the explicit expressions for s_k, R_k, and I_k, and other relevant quantities or functions. We start from

$$s_k=2\left(J_xe^{ik\cdot\hat{\mathbf{n}}_1}+J_ye^{ik\cdot\hat{\mathbf{n}}_2}+J_z\right)$$

Then, if we express the momentum state in terms of $\mathbf{k}=k_x\hat{\mathbf{e}}_x+k_y\hat{\mathbf{e}}_2$, then

$$\left.\begin{aligned}|s_k|&=2\left[J_x^2+J_y^2+J_z^2+2J_xJ_y\cos\left(k_x\right)+2J_xJ_z\cos\left(\frac{1}{2}k_x+\frac{\sqrt{3}}{2}k_y\right)+2J_yJ_z\cos\left(\frac{1}{2}k_x-\frac{\sqrt{3}}{2}k_y\right)\right]^{1/2}\\ R_k&=2\left[J_x\cos\left(\frac{1}{2}k_x+\frac{\sqrt{3}}{2}k_y\right)+J_y\cos\left(\frac{1}{2}k_x-\frac{\sqrt{3}}{2}k_y\right)+J_z\right]\\ I_k&=2\left[J_x\sin\left(\frac{1}{2}k_x+\frac{\sqrt{3}}{2}k_y\right)-J_y\sin\left(\frac{1}{2}k_x-\frac{\sqrt{3}}{2}k_y\right)\right]\end{aligned}\right\}\tag{C.4}$$

Or, expressed in terms of reciprocal basis vectors $\mathbf{k}=k_1\hat{\mathbf{b}}_1+k_2\hat{\mathbf{b}}_2$, then

$$\left.\begin{aligned}|s_k|&=2\left[J_x^2+J_y^2+J_z^2+2J_xJ_y\cos\left(2\pi k_1\right)+2J_xJ_z\cos\left(2\pi k_2\right)+2J_yJ_z\cos\left(2\pi(k_1-k_2)\right)\right]^{1/2}\\ R_k&=2\left[J_x\cos\left(2\pi k_2\right)+J_y\cos\left(2\pi(k_1-k_2)\right)+J_z\right]\\ I_k&=2\left[J_x\sin\left(2\pi k_2\right)-J_y\sin\left(2\pi(k_1-k_2)\right)\right]\end{aligned}\right\}\tag{C.5}$$

Again, these results are general and are valid even in a finite-size system.

With these relations, we can start to compute the c-Majorana correlation functions. For AA-pairings, the correlation function is,

$$\begin{aligned}\langle c_{\mathbf{x}A}c_{\mathbf{x}'A}\rangle&=\frac{1}{N/2}\sum_{\mathbf{k}}\left(\langle\gamma_{\mathbf{k}}\gamma_{-\mathbf{k}}\rangle+\langle\gamma_{-\mathbf{k}}^\dagger\gamma_k^\dagger\rangle+\langle\gamma_{\mathbf{k}}\gamma_k^\dagger\rangle+\langle\gamma_{-\mathbf{k}}^\dagger\gamma_{-\mathbf{k}}\rangle\right)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}\\&=\frac{1}{N/2}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})}=\delta_{\mathbf{x}\mathbf{x}'}\end{aligned}$$

For AB-pairings, a similar calculation gives

$$\begin{aligned}\langle c_{\mathbf{x}A}c_{\mathbf{x}'B}\rangle &= \frac{i}{N/2}\sum_{\mathbf{k}}\left(\langle-\gamma_{\mathbf{k}}\gamma_{-\mathbf{k}}\rangle+\langle\gamma^{\dagger}_{-\mathbf{k}}\gamma^{\dagger}_{k}\rangle+\langle\gamma_{\mathbf{k}}\gamma^{\dagger}_{k}\rangle-\langle\gamma^{\dagger}_{-\mathbf{k}}\gamma_{-\mathbf{k}}\rangle\right)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}\\ &= \frac{i}{N/2}\sum_{\mathbf{k}}\left(\frac{R_{\mathbf{k}}}{|s_{\mathbf{k}}|}-i\frac{I_k}{|s_{\mathbf{k}}|}\right)e^{i\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})}=\frac{i}{N/2}\sum_{\mathbf{k}}\frac{s^{*}_{\mathbf{k}}}{|s_{\mathbf{k}}|}e^{i\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})}\end{aligned}$$

This result is purely imaginary and can be further simplified,

$$\begin{aligned}\langle -ic_{\mathbf{x}A}c_{\mathbf{x}'B}\rangle &= \frac{1}{N/2}\sum_{\mathbf{k}}\cos\left[\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})-\theta_{\mathbf{k}}\right]\\ &= \frac{1}{N/2}\sum_{\mathbf{k}}\frac{1}{|s_{\mathbf{k}}|}\left[R_{\mathbf{k}}\cos\left(\mathbf{k}\cdot\delta\mathbf{x}\right)+I_{\mathbf{k}}\sin\left(\mathbf{k}\cdot\delta\mathbf{x}\right)\right]\end{aligned}\tag{C.6}$$

where $s_{\mathbf{k}}=|s_{\mathbf{k}}|\,e^{i\theta_{\mathbf{k}}}$. Notice that the above formulation is general and can be used directly in the case of finite systems.

C.3 Two-Points Correlation Functions with an External Magnetic Field

In the isotropic limit, applying an external magnetic field makes the Kitaev model equivalent to a Haldane honeycomb model [7] in the Majorana fermion representation. In general, the correlation functions of AA and BB pairings are finite. However, since the $\mathbb{Z}_2$ gauge fields are still c-numbers, the ground state still exhibits the pattern that only string-like operators bear finite expectation values. The only difference is that, in the situation with a magnetic field, the N.N.N. coupling contributes to the correlation functions of longer strings, and strings with both endpoints of the same type have finite expectation values in general.

If a magnetic field is applied to the system, then the elementary excitations are given in Eq. (F.9), which satisfy

$$\begin{cases}\langle a^{\dagger}_{\mathbf{k}}a_{\mathbf{k}'}\rangle\delta_{\mathbf{k}\mathbf{k}'}\\ \langle a_{\mathbf{k}}a_{\mathbf{k}'}\rangle=\langle a_{\mathbf{k}}a^{\dagger}_{\mathbf{k}'}\rangle=0\end{cases}$$

Therefore, the γ-fermions can be represented in terms of the elementary excitations as follows,

$$\begin{pmatrix}\gamma_{\mathbf{k}}\\ \gamma^{\dagger}_{-\mathbf{k}}\end{pmatrix}=\begin{pmatrix}u_{\mathbf{k}}a_{\mathbf{k}}+e^{-i\varphi_{\mathbf{k}}}v_k a^{\dagger}_{-\mathbf{k}}\\ e^{i\varphi_{\mathbf{k}}}v_k a_{\mathbf{k}}+u_{\mathbf{k}}a^{\dagger}_{-\mathbf{k}}\end{pmatrix}.$$

$$\begin{cases}\langle\gamma^{\dagger}_{\mathbf{k}}\gamma_{\mathbf{k}'}\rangle=u^2_{\mathbf{k}}\delta_{\mathbf{k}\mathbf{k}'}=\dfrac{1}{2}\left(1+\dfrac{R_{\mathbf{k}}}{\epsilon(\mathbf{k})}\right)\delta_{\mathbf{k}\mathbf{k}'}\\ \langle\gamma_{\mathbf{k}}\gamma^{\dagger}_{\mathbf{k}'}\rangle=\dfrac{1}{2}\left(1-\dfrac{R_{\mathbf{k}}}{\epsilon(\mathbf{k})}\right)\delta_{\mathbf{k}\mathbf{k}'}\\ \langle\gamma_{\mathbf{k}}\gamma_{\mathbf{k}'}\rangle=e^{-i\varphi_{\mathbf{k}}}v_{\mathbf{k}}u_{-\mathbf{k}}\delta_{\mathbf{k},-\mathbf{k}'}=\dfrac{\Delta_{\mathbf{k}}-iI_{\mathbf{k}}}{2\epsilon(\mathbf{k})}\delta_{\mathbf{k},-\mathbf{k}'}\\ \langle\gamma^{\dagger}_{\mathbf{k}}\gamma^{\dagger}_{\mathbf{k}'}\rangle=\langle\gamma_{\mathbf{k}'}\gamma_{\mathbf{k}}\rangle^{*}=-\dfrac{\Delta_{\mathbf{k}}+iI_{\mathbf{k}}}{2\epsilon(\mathbf{k})}\delta_{\mathbf{k},-\mathbf{k}'}\end{cases}$$

Note that $u_{-\mathbf{k}}=u_{\mathbf{k}}$.

There are two types of two-point correlation functions. The first one is the pairing in the same sublattice.

$$\begin{aligned}\langle -ic_{\mathbf{x}A}c_{\mathbf{x}'A}\rangle &= \frac{-i}{N/2}\sum_{\mathbf{k}}\left(\langle\gamma_{\mathbf{k}}\gamma_{-\mathbf{k}}\rangle + \langle\gamma^\dagger_{-\mathbf{k}}\gamma^\dagger_{k}\rangle + \langle\gamma_{\mathbf{k}}\gamma^\dagger_{\mathbf{k}}\rangle + \langle\gamma^\dagger_{-\mathbf{k}}\gamma_{-\mathbf{k}}\rangle\right)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \\ &= \frac{-i}{N/2}\sum_{\mathbf{k}}\left(1+\frac{\Delta_{\mathbf{k}}}{\epsilon(\mathbf{k})}\right)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')}\end{aligned}$$

$$\langle -ic_{\mathbf{x}A}c_{\mathbf{x}'A}\rangle = -i\delta_{\mathbf{x}\mathbf{x}'} + \frac{1}{N/2}\sum_{\mathbf{k}}\frac{\Delta_{\mathbf{k}}}{\epsilon(\mathbf{k})}\sin\left(\mathbf{k}\cdot\delta\mathbf{x}\right) \tag{C.7}$$

$$\begin{aligned}\langle -ic_{\mathbf{x}A}c_{\mathbf{x}'B}\rangle &= \frac{1}{N/2}\sum_{\mathbf{k}}\left(-\langle\gamma_{\mathbf{k}}\gamma_{-\mathbf{k}}\rangle + \langle\gamma^\dagger_{-\mathbf{k}}\gamma^\dagger_{k}\rangle + \langle\gamma_{\mathbf{k}}\gamma^\dagger_{k}\rangle - \langle\gamma^\dagger_{-\mathbf{k}}\gamma_{-\mathbf{k}}\rangle\right)e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{x}')} \\ &= \frac{1}{N/2}\sum_{\mathbf{k}}\left(\frac{R_{\mathbf{k}}}{\epsilon(\mathbf{k})} + i\frac{I_k}{\epsilon(\mathbf{k})}\right)e^{i\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})} \\ &= \frac{1}{N/2}\sum_{\mathbf{k}}\frac{|s_k|}{\epsilon(\mathbf{k})}e^{i\mathbf{k}\cdot(\mathbf{x}'-\mathbf{x})+i\theta_{\mathbf{k}}} \in \mathbb{R}\end{aligned}$$

$$\langle -ic_{\mathbf{x}A}c_{\mathbf{x}'B}\rangle = \frac{1}{N/2}\sum_{\mathbf{k}}\left[\frac{R_{\mathbf{k}}}{\epsilon(\mathbf{k})}\cos\left(\mathbf{k}\cdot\delta\mathbf{x}\right) - \frac{I_{\mathbf{k}}}{\epsilon(\mathbf{k})}\sin\left(\mathbf{k}\cdot\delta\mathbf{x}\right)\right] \tag{C.8}$$

C.4 Nearest Neighbor Correlation Functions

In the toric code limit, the dominant correlation functions are those of nearest-neighbor lattice sites along the strong coupling direction. Without loss of generality, assume the dominant coupling is J_z, so that $J_x = J_y = J_z\delta$ can be treated as perturbations. In this case, the term $|s_k|$ appearing in the denominator in the integrand is

$$\begin{aligned}\frac{|s_k|^2}{4J_z^2} &= 1 + 2\delta^2 + 2\delta^2\cos\left(\mathbf{k}\cdot(\hat{\mathbf{n}}_1-\hat{\mathbf{n}}_2)\right) + 2\delta\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right) + 2\delta\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right) \\ &= 1 + 2\delta^2\left[1+\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right)\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right) + \sin\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right)\sin\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right)\right] \\ &\quad + 2\delta\left[\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right) + \cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right)\right] \\ &= \left[1+\delta\cos(\mathbf{k}\cdot\hat{\mathbf{n}}_1) + \delta\cos(\mathbf{k}\cdot\hat{\mathbf{n}}_2)\right]^2 + \left[\delta\sin(\mathbf{k}\cdot\hat{\mathbf{n}}_1) + \delta\sin(\mathbf{k}\cdot\hat{\mathbf{n}}_2)\right]^2 \\ &\equiv (1+\delta u)^2 + \delta^2 v^2\end{aligned} \tag{C.9}$$

$$\frac{R_k}{2J_z} = 1 + \delta\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right) + \delta\cos\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right) \equiv 1 + \delta u \tag{C.10}$$

$$\delta = \frac{\alpha/3}{1-2\alpha/3} \approx \frac{\alpha}{3} + \frac{2\alpha^2}{9}$$

$$
\begin{aligned}
\langle -ic_{\mathbf{x}A}c_{\mathbf{x}B}\rangle &= \frac{1}{N/2}\sum_{\mathbf{k}}\frac{R_{\mathbf{k}}}{\epsilon(\mathbf{k})} = \frac{1}{N/2}\sum_{\mathbf{k}}\frac{R_{\mathbf{k}}}{|s_k|} \\
&= \frac{1}{N/2}\sum_{\mathbf{k}}\frac{1+\delta u}{\sqrt{(1+\delta u)^2+\delta^2 v^2}} \\
&= \frac{1}{N/2}\sum_{k}\left[1+\left(\frac{\delta v}{1+\delta u}\right)^2\right]^{-1/2} \\
&\approx \frac{1}{N/2}\sum_{k}\left[1-\frac{1}{2}\delta^2\left(\sin\left(\mathbf{k}\cdot\hat{\mathbf{n}}_1\right)+\sin\left(\mathbf{k}\cdot\hat{\mathbf{n}}_2\right)\right)^2\right] \\
&= 1-\frac{1}{2}\delta^2+\mathcal{O}(\delta^3)
\end{aligned}
$$

$$
\langle -ic_{\mathbf{x}A}c_{\mathbf{x}B}\rangle \approx 1-\frac{\alpha^2}{18}+\mathcal{O}(\alpha^3) \tag{C.11}
$$

$$
\det \mathsf{C}_{\mathfrak{D}} \approx \left(1-\frac{\alpha^2}{18}\right)^{2N_d^{\mathfrak{D}}}
$$

$$
\begin{aligned}
\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4 &= \sum_{[\mathfrak{D}]\subset A}\left[\det(\mathsf{C}_{\mathfrak{D}})\right]^2 \approx \sum_{n=0}^{N_d^A}\left(1-\frac{\alpha^2}{18}\right)^{4n} \\
&= \frac{\left(1-\frac{\alpha^2}{18}\right)^{4\left(N_d^A+1\right)}-1}{\left(1-\frac{\alpha^2}{18}\right)^4-1} \\
&\approx \frac{4\left(N_d^A+1\right)-\binom{4\left(N_d^A+1\right)}{2}\frac{\alpha^2}{18}}{4-\binom{4}{2}\frac{\alpha^2}{18}}
\end{aligned}
$$

$$
-\ln\left(\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^4\right)-\ln\det\left(\mathbb{1}+\mathsf{C}_A\right).
$$

C.5 Two-Points Correlation Functions in a Finite-Size System

In the following discussions, we consider the lattice geometry illustrated in Fig. 4.c. In this situation, there are three methods for labeling the position of lattice sites.

- The first method is to assign a number to each lattice site one by one. The label is denoted by n in this note.
- In the lattice geometry illustrated in Fig. 4.c, the lattice sites can be sorted into layers labeled by l, and the horizontal position of a lattice site in each layer is labeled by s, standing for *site*.

For example, the position of site number 7 is $(l, s) = (1, 3)$, and the position of site number 12 is $(l, s) = (3, 0)$. The integer s can be further decomposed as

$$s = 2s_q + s_r$$

where $s_q \in \mathbb{N}$ and $s_r = 0 \vee 1$ is the remainder.

- With the Bravais lattice site expressed as $\mathbf{x} = x\hat{\mathbf{a}}_1 + y\hat{\mathbf{a}}_2$, each lattice site corresponds to a unique pair $(\mathbf{x}, \mu)$, where $\mu = 0, 1$, with 0 representing sublattice B and 1 representing A, respectively.
- The following equations describe the relationships among the three labeling methods listed above.

$$\begin{cases} n = s + lN_h \\ \mu = n \mod 2 \\ x = s_q + s_r \\ y = l - s_r \end{cases}$$

With this definition, the factor $\mathbf{k} \cdot \delta\mathbf{x}$ in Eq. (C.6) can be written as

$$\mathbf{k} \cdot \delta\mathbf{x} = \pi \left(n_1 \delta x + \frac{n_4}{2} \delta y \right)$$

where n_1 and n_2 are defined by the expression Eq. (C.3) of momentum states, and δx and δy are components of the position $\mathbf{x}$ represented in terms of lattice unit vectors $\mathbf{x} = \delta x\hat{\mathbf{a}}_1 + \delta y\hat{\mathbf{a}}_2$. Using this representation, the summation $\sum_{\mathbf{k}}$ in Eq. (C.6) can be written as $\sum_{n_1,n_2}$. The only difference is that the momentum is now represented in terms of integers n_1 and n_2.

C.6 String Correlation Functions

$$\prod_{\langle ij \rangle \in \mathfrak{D}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C} \in \mathrm{St}(\mathfrak{D})} \left(\prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \right)$$

We have required $\mathcal{C} \in \mathbf{G}$. That is to say, each string $\mathcal{C}$ connects a type-A and a type-B endpoint.

$$\begin{aligned} \prod_{\langle ij \rangle \in \mathcal{C}} \sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}} \to \prod_{\langle ij \rangle \in \mathcal{C}} (-i) c_i c_j &= (-i)^{|\mathcal{C}|} (-1)^{(|\mathcal{C}|-1)/2} c_{a_\mathcal{C}} c_{b_\mathcal{C}} \\ &= -i(-1)^{|\mathcal{C}|(|\mathcal{C}|+1)/2} (-1)^{(|\mathcal{C}|+1)/2} c_{a_\mathcal{C}} c_{b_\mathcal{C}} \\ &= -i(-1)^{(|\mathcal{C}|+1)^2/2} c_{a_\mathcal{C}} c_{b_\mathcal{C}} \end{aligned}$$

In the derivation, we have used the identity $i^n = (-i)^n = (-1)^{n(n+1)/2}$ if n is an even integer. The sign factor $(-1)^{(|\mathcal{C}|-1)/2}$ is due to reversing the direction of dimers to make them point from B to A. Notice that in each dimer $\sigma_i^{\alpha_{ij}} \sigma_j^{\alpha_{ij}}$, the former index i represents the lattice site in the A-sublattice, while the latter one j stands for the B-sublattice, and the direction of the string $\mathcal{C}$ is always from its type-A endpoint to the type-B one. At the same time, because the two endpoints of each string are in different sublattices, considering the honeycomb lattice geometry, the length $|\mathcal{C}|$ of each string $\mathcal{C}$ must be an odd number. With this property, the factor $(|\mathcal{C}|+1)^2$ that appears

in the exponent of the sign factor must be a multiple of 4, making the sign $(-1)^{(|\mathcal{C}|+1)^2/2}$ trivial. Therefore, we eventually obtain the correspondence between a string operator and its Majorana representation, which depends only on the endpoints $a_{\mathcal{C}}$ and $b_{\mathcal{C}}$ if $\mathcal{C}$ is a *good* string

$$\prod_{\langle ij\rangle\in\mathcal{C}\subset\mathbf{G}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \to -ic_{a_{\mathcal{C}}}c_{b_{\mathcal{C}}} \tag{C.12}$$

$$\prod_{\langle ij\rangle\in\mathfrak{D}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} = \prod_{\mathcal{C}\in\mathrm{St}(\mathfrak{D})}\left(\prod_{\langle ij\rangle\in\mathcal{C}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}}\right) \to \prod_{\mathcal{C}\in\mathrm{St}(\mathfrak{D})}(-ic_{a_{\mathcal{C}}}c_{b_{\mathcal{C}}}) \tag{C.13}$$

$$\begin{aligned}\left\langle\prod_{\langle ij\rangle\in\mathfrak{D}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}}\right\rangle &\to \left\langle\prod_{\mathcal{C}\in\mathrm{St}(\mathfrak{D})}(-ic_{a_{\mathcal{C}}}c_{b_{\mathcal{C}}})\right\rangle_M \\ &= \begin{vmatrix} \langle -ic_{a_1}c_{b_1}\rangle_M & \langle -ic_{a_1}c_{b_2}\rangle_M & \langle -ic_{a_1}c_{b_3}\rangle_M & \cdots \\ \langle -ic_{a_2}c_{b_1}\rangle_M & \langle -ic_{a_2}c_{b_2}\rangle_M & \langle -ic_{a_2}c_{b_3}\rangle_M & \cdots \\ \langle -ic_{a_3}c_{b_1}\rangle_M & \langle -ic_{a_3}c_{b_2}\rangle_M & \langle -ic_{a_3}c_{b_3}\rangle_M & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix}\end{aligned} \tag{C.14}$$

The result can also be expressed in terms of the Pfaffian[3] [4].

$$\mathfrak{C}_{\mathfrak{D}} = \left\langle\prod_{\mathcal{C}\in\mathrm{St}(\mathfrak{D})}(-ic_{a_{\mathcal{C}}}c_{b_{\mathcal{C}}})\right\rangle_M = \mathrm{Pf}(\mathsf{C}_{\mathfrak{D}}) \tag{C.15}$$

where the skew-symmetric matrix $\mathsf{C}_{\mathfrak{D}}$ is defined as follows

$$[\mathsf{C}_{\mathfrak{D}}]_{ij} = \begin{cases} 0 & ; \quad i=j \\ \langle -ic_ic_j\rangle_M & ; \quad i\neq j, \quad i,j\in \mathrm{EP}(\mathfrak{D}) \end{cases} \tag{C.16}$$

In the above definition of $A_{\mathfrak{D}}$, lattice sites i and j may belong to different sublattices.

Theorem C.1. *For any $N\times N$ skew-symmetric matrix A, the equation below holds [4]*

$$\det(\mathbb{1}+A) = \sum_{S\subset[N]}\det(A_S) = \sum_{S\subset[N]}[\mathrm{Pf}(A_S)]^2 .$$

In the above equation, $[N]\equiv\{1,2,\cdots,N\}$ denotes the set of integers from 1 to N, and A_S represents a skew-symmetric matrix whose rows and columns are those of A at indices from S.

Proof. The matrix A can be expressed as an assembly of column vectors $A=(\mathbf{a}_1,\mathbf{a}_2,\cdots,\mathbf{a}_N)$. Similarly, the identity matrix can also be written in terms of unit vectors $\mathbb{1}=(\mathbf{e}_1,\mathbf{e}_2,\cdots,\mathbf{e}_N)$. With these expressions, the determinant can be expressed in terms of column vectors. Due to this fact and the linearity of the determinant, the function $\det(\mathbb{1}+A)$ can be expanded.

$$\begin{aligned}\det(\mathbb{1}+A) &= \det(\mathbf{e}_1+\mathbf{a}_1,\mathbf{e}_2+\mathbf{a}_2,\cdots\mathbf{e}_N+\mathbf{a}_N) \\ &= \det(\mathbf{e}_1,\mathbf{e}_2+\mathbf{a}_2,\cdots\mathbf{e}_N+\mathbf{a}_N)+\det(\mathbf{a}_1,\mathbf{e}_2+\mathbf{a}_2,\cdots\mathbf{e}_N+\mathbf{a}_N) \\ &= \sum_{S\in[N]}\det\left(\cdots,\mathbf{a}_{S(1)},\cdots,\mathbf{a}_{S(2)},\cdots,\mathbf{a}_{S(\mathrm{f.})},\cdots\right)\end{aligned}$$

[3] For a 4×4 skew-symmetric matrix A, the Pffafian reads,

$$\mathrm{Pf}(A) = A_{12}A_{34}-A_{13}A_{24}+A_{14}A_{23}.$$

The dots in the last line of the above equations are filled with unit vectors, and $S(i)$ represents the i-th element of S. Each determinant in the last equation is equal to the determinant of A_S since the matrix is left with only the matrix elements of A_S that can contribute to the determinant. The unit vectors serve as indicators of the positions at which the rows and columns are to be eliminated.

$$\det\left(\cdots, \mathbf{a}_{S(1)}, \cdots, \mathbf{a}_{S(2)}, \cdots, \mathbf{a}_{S(\mathrm{f.})}, \cdots\right) = \det(A_S).$$

Substitution of this result into the previous equation completes the proof. ■

In the following, we prove the normalization condition Eq. (D.3) using the correlation matrix technique discussed earlier. To this end, we first observe that the Majorana correlation matrix Γ_R consisting of Majorana fermions living in R, defined as

$$[\Gamma_R]_{ij} = \frac{1}{2}\langle c_i c_j \rangle \ , \qquad i, j \in R,$$

can be expressed in terms of $A_{\mathfrak{D}_Z}$ as follows,

$$\Gamma_R = \frac{1}{2}\mathbb{1} + \frac{i}{2}\mathsf{C}_R.$$

If the region R is chosen as the entire system Λ, then we have the equation below,

$$\mathbb{1} + i\mathsf{C}_{\mathfrak{D}_Z} = 2\Gamma_\Lambda = 2U\mathcal{G}_\Lambda U^\dagger.$$

In the above equation, U is the unitary Bogoliubov transformation, and the matrix $\mathcal{G}$ defined as $\mathcal{G}_{i\sigma\aleph, j\sigma'\aleph'} \equiv \langle b_{i\sigma\aleph} b_{j\sigma'\overline{\aleph'}} \rangle$ is the general Green's function matrix in the particle-hole formalism. Fermions b are the elementary excitations that diagonalize the Hamiltonian. With the above equation, the square of the matrix $\mathsf{C}_{\mathfrak{D}_Z}$ can be represented in terms of the Green's function matrix

$$\mathsf{C}^2_{\mathfrak{D}_Z} = -U\left(\mathbb{1} - 4\mathcal{G}_\Lambda + 4\mathcal{G}^2_\Lambda\right)U^\dagger. \tag{C.17}$$

Here, we choose the state as the ground state of the free-fermion system. In this case, the Green's function matrix is a projector, i.e., $\mathcal{G}^2_\Lambda = \mathcal{G}_\Lambda$. The following derivation demonstrates the proof.

$$\begin{aligned}
[\mathcal{G}^2_\Lambda]_{i\sigma\aleph, j\sigma'\aleph'} &= \sum_{ks\beth} \langle b_{i\sigma\aleph} b_{ks\bar{\beth}} \rangle \langle b_{ks\beth} b_{j\sigma'\overline{\aleph'}} \rangle \\
&= \delta_{ij}\delta_{\sigma\sigma'}\delta_{\aleph\aleph'} \sum_{l\in F} \left[n_l^2 + (1-n_l)^2\right] = \delta_{ij}\delta_{\sigma\sigma'}\delta_{\aleph\aleph'} \sum_{l\in F} \\
&= [\mathcal{G}_\Lambda]_{i\sigma\aleph, j\sigma'\aleph'},
\end{aligned}$$

where F denotes the Fermi volume of the ground state. Due to this result, Eq. (C.17) reduces to

$$\mathsf{C}^2_{\mathfrak{D}_Z} = -\mathbb{1}.$$

The above equation implies that the skew-symmetric matrix $\mathsf{C}_{\mathfrak{D}_Z}$ can be represented in the following standard form,

$$\mathsf{C}_{\mathfrak{D}_Z} = O\left[\bigoplus_{n=1}^{N/2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right] O^T,$$

which means the eigenvalues of $\mathsf{C}_{\mathfrak{D}_Z}$ are $\{+i,-i\}^{\otimes N/2}$. Together, we have two different approaches to calculate $\det(\mathbb{1}+\mathsf{C}_{\mathfrak{D}_Z})$. The first approach is based on Theorem C.1, and the other is exploiting the eigenvalues of $\mathsf{C}_{\mathfrak{D}_Z}$ implied by its standard form.

$$\begin{cases} \det(\mathbb{1}+\mathsf{C}_{\mathfrak{D}_Z}) = \sum\limits_{\mathrm{EP}(\mathfrak{D})} \left[\mathrm{Pf}(\mathsf{C}_{\mathfrak{D}})\right]^2 = \sum\limits_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}}^2 \\ \det(\mathbb{1}+\mathsf{C}_{\mathfrak{D}_Z}) = \prod\limits_{k=1}^{N/2}(1+i)(1-i) = 2^{N/2} = 2^{N-N_p} \end{cases}$$

Since the two approaches must lead to the same result, we conclude that the following equation holds for the correlation functions in a pure Kitaev model

$$\sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}}^2 = 2^{N-N_p} \tag{C.18}$$

which proves Eq. (D.3) and further implies that the density matrix proposed in [2] is indeed a pure state.

In general, the length of the string $|\mathcal{C}|$ can be even since a string in TRodd can have a finite expectation value if an external magnetic field is applied. In this case, the above formulas become

$$\begin{aligned} \prod_{\langle ij\rangle\in\mathcal{C}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \to \prod_{\langle ij\rangle\in\mathcal{C}} (-i)c_ic_j &= (-i)^{|\mathcal{C}|}(-1)^{|\mathcal{C}|/2}c_sc_f \\ &= (-1)^{|\mathcal{C}|(|\mathcal{C}|-1)/2}(-1)^{|\mathcal{C}|/2}c_sc_f \\ &= (-1)^{|\mathcal{C}|^2/2}c_sc_f = c_sc_f \end{aligned}$$

Therefore, in general, the multi-point correlation function reads

$$\left\langle \prod_{\langle ij\rangle\in\mathfrak{D}} \sigma_i^{\alpha_{ij}}\sigma_j^{\alpha_{ij}} \right\rangle \to i^{\#\text{ of }\mathbf{TRodd}\text{ strings}} \left\langle \prod_{\mathcal{C}\in\mathrm{St}(\mathfrak{D})} (-ic_{s_\mathcal{C}}c_{f_\mathcal{C}}) \right\rangle_M$$

Or, equivalently

$$\mathfrak{C}_{\mathfrak{D}}^2 = \langle\Sigma_{\mathfrak{D}}\rangle^2 = \eta_{\mathfrak{D}}\mathrm{Pf}\left(\mathsf{C}_{\mathfrak{D}}\right)^2, \tag{C.19}$$

which means

$$\sum_{[\mathfrak{D}]} \eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2 = \sum_{[\mathfrak{D}]} \mathrm{Pf}(\mathsf{C}_{\mathfrak{D}})^2 = \det\left(\mathbb{1}+\mathsf{C}_{\mathfrak{D}_Z}\right) = 2^{N-N_p}$$

D Sum Rules for Correlation Functions

In this appendix, we prove the most general sum rule for the correlation functions at the ground state of the Kitaev model, which applies to both the pure Kitaev model and the situation with an external magnetic field (general Kitaev-like models). First, the density matrix of the ground state reads

$$\rho = \frac{1}{2^{N-N_p}}\hat{\Pi}\sum_{[\mathfrak{D}]} \mathfrak{C}_{\mathfrak{D}}\Sigma_{\mathfrak{D}} \tag{D.1}$$

The coefficients are defined as $\mathfrak{C}_{\mathfrak{D}} \equiv \eta_{\mathfrak{D}}\langle\Sigma_{\mathfrak{D}}\rangle$, where $\eta_{\mathfrak{D}}$ is the sign obtained by squaring the string operator $\Sigma_{\mathfrak{D}}^2 = \eta_{\mathfrak{D}}\mathbb{1}$ because not every string squares to one. The projector $\hat{\Pi}$ is defined as follows,

$$\hat{\Pi} = \left(\frac{\mathbb{1}+\tilde{W}_v}{2}\right)\left(\frac{\mathbb{1}+\tilde{W}_h}{2}\right)\prod_p\left(\frac{\mathbb{1}+W_p}{2}\right).$$

It projects the state to the flux-free subspace in the topological sector labeled by the expectation value of the non-contractible Wilson loops $(\tilde{W}_v, \tilde{W}_h) = (+,+)$.

A fundamental condition for a density matrix to be a pure state is that it must square to one. Directly taking the square, we obtain

$$\begin{aligned}
\rho^2 &= \frac{1}{2^{2N-2N_p}}\hat{\Pi}\sum_{[\mathfrak{D}]}\sum_{[\mathfrak{D}']}\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}'}\varphi(\mathfrak{D},\mathfrak{D}')\Sigma_{\mathfrak{D}\ominus\mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}}\hat{\Pi}\sum_{[\mathfrak{D}]}\sum_{[\mathfrak{D}']}\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}'}\frac{1}{2}\left[\varphi(\mathfrak{D},\mathfrak{D}')+\varphi(\mathfrak{D}',\mathfrak{D})\right]\Sigma_{\mathfrak{D}\ominus\mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}}\hat{\Pi}\sum_{[\mathfrak{D}]}\sum_{[\mathfrak{D}']}\mathfrak{C}_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}'}\vartheta(\mathfrak{D},\mathfrak{D}')\Sigma_{\mathfrak{D}\ominus\mathfrak{D}'} \\
&= \frac{1}{2^{2N-2N_p}}\hat{\Pi}\sum_{[\mathfrak{D}^c]}\sum_{[\mathfrak{D}']}\mathfrak{C}_{\mathfrak{D}^c\ominus\mathfrak{D}'}\mathfrak{C}_{\mathfrak{D}'}\vartheta(\mathfrak{D}^c\ominus\mathfrak{D}',\mathfrak{D}')\Sigma_{\mathfrak{D}^c} \\
&= \frac{1}{2^{N-N_p}}\hat{\Pi}\sum_{[\mathfrak{D}]}\left(\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}']}\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{D}'}\mathfrak{C}_{\mathfrak{D}'}\eta_{\mathfrak{D}'}\vartheta(\mathfrak{D},\mathfrak{D}')\right)\Sigma_{\mathfrak{D}}
\end{aligned}$$

Comparing this result with Eq. (D.1), we obtain the *strong condition*,

$$\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}']}\eta_{\mathfrak{D}'}\vartheta(\mathfrak{D},\mathfrak{D}')\mathfrak{C}_{\mathfrak{D}\ominus\mathfrak{D}'}\mathfrak{C}_{\mathfrak{D}'} = \mathfrak{C}_{\mathfrak{D}}. \tag{D.2}$$

The symmetric phase factor $\vartheta(\mathfrak{D},\mathfrak{D}')$ is defined by symmetrizing the phase factor $\varphi(\mathfrak{D},\mathfrak{D}')$ defined by the projective representation of the edge configuration algebra.

$$\vartheta(\mathfrak{D},\mathfrak{D}') \equiv \frac{1}{2}\left[1+\eta_{\mathfrak{D}}\eta_{\mathfrak{D}'}\eta_{\mathfrak{D}'\ominus\mathfrak{D}}\right]\varphi(\mathfrak{D},\mathfrak{D}') = \vartheta(\mathfrak{D}',\mathfrak{D}) = 0 \text{ or } \pm 1$$

Since the function φ has the following property,

$$\varphi(\mathfrak{D}\ominus\mathfrak{D}',\mathfrak{D}') = \eta_{\mathfrak{D}}\eta_{\mathfrak{D}\ominus\mathfrak{D}'}\varphi(\mathfrak{D}',\mathfrak{D}) = \eta_{\mathfrak{D}'}\varphi(\mathfrak{D},\mathfrak{D}'),$$

The symmetric phase factor obeys a similar relation

$$\vartheta(\mathfrak{D}'\ominus\mathfrak{D},\mathfrak{D}) = \frac{1}{2}\left[1+\eta_{\mathfrak{D}\ominus\mathfrak{D}'}\eta_{\mathfrak{D}'}\eta_{\mathfrak{D}}\right]\varphi(\mathfrak{D}'\ominus\mathfrak{D},\mathfrak{D}) = \eta_{\mathfrak{D}}\vartheta(\mathfrak{D}',\mathfrak{D}).$$

Taking $\mathfrak{D}' = \varnothing$ and the property $\varphi(\mathfrak{D},\mathfrak{D}) = \eta_{\mathfrak{D}}$, we have

$$\vartheta\left(\mathfrak{D},\mathfrak{D}\right) = \frac{1}{2}\left[1+\eta_{\mathfrak{D}}\eta_{\mathfrak{D}}\eta_{\mathfrak{D}\ominus\mathfrak{D}}\right]\eta_{\mathfrak{D}} = \eta_{\mathfrak{D}}.$$

With this property and taking $\mathfrak{D} = \varnothing$, the strong condition reduces to the *weak condition*

$$\frac{1}{2^{N-N_p}}\sum_{[\mathfrak{D}]}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2 = 1. \tag{D.3}$$

E Rényi Entropy

The Rényi entropy (entanglement entropy) part can be calculated numerically by diagonalizing the correlation matrix of the Majorana fermions in the subsystem A.

$$\sum_{[\mathfrak{D}]\subset A}\eta_{\mathfrak{D}}\mathfrak{C}_{\mathfrak{D}}^2=\sum_{\mathrm{EP}(\mathfrak{D})\subset R}\left[\mathrm{Pf}(\mathsf{C}_{\mathfrak{D}})\right]^2=\sum_{\mathrm{EP}(\mathfrak{D})\subset R}\det(\mathsf{C}_{\mathfrak{D}})=\det\left(\mathbb{1}+\mathsf{C}_A\right)$$

$$\Gamma_A=\frac{1}{2}\mathbb{1}+\frac{i}{2}\mathsf{C}_A$$

Since the spectrum of the correlation matrix Γ_A is the single-particle entanglement spectrum

$$\gamma_{k,\pm}=\frac{1\pm\nu_k}{2}\,,\qquad k=1,2,\cdots,\frac{N^A}{2}$$

of the reduced density matrix ρ_A of the subregion A, the eigenvalues of C_A are

$$\pm i\frac{\nu_k}{2}\,,\qquad k=1,2,\cdots,\frac{N^A}{2}. \tag{E.1}$$

Therefore, the 2-Rényi entropy reduces to

$$\begin{aligned}-\ln\left(\frac{1}{2^{N^A-N_p^A}}\sum_{[\mathfrak{D}]\subset A}\mathfrak{C}_{\mathfrak{D}}^2\right)&=-\ln\det\left(\mathbb{1}+\mathsf{C}_A\right)+\left(N^A-N_p^A\right)\ln 2\\&=-\ln\left[\prod_{k=1}^{N^A/2}\frac{1}{2}\left(1+i\nu_k\right)\left(1-i\nu_k\right)\right]+\left(\frac{N^A}{2}-N_p^A\right)\ln 2\\&=\underbrace{-\sum_{k=1}^{N^A/2}\ln\left(\frac{1+\nu_k^2}{2}\right)}_{\text{matter}}+\underbrace{\left(\frac{1}{2}\left|\partial A\right|-1\right)\ln 2}_{\text{gauge}}\end{aligned}$$

F Exactly Solving for the Ground State

According to equation (29) of Ref. [8], we can diagonalize the Kitaev Hamiltonian in the momentum space if the gauge is fixed to be $u_{ij}=+1\forall\ (ij)$,

$$\hat{H}=\sum_{\mathbf{k}}\Psi_k^\dagger U(k)\Psi_k=\frac{1}{2}\sum_{\mathbf{k}}\left[is_{\mathbf{k}}d_{k,A}^\dagger d_{k,B}-is_{\mathbf{k}}^*d_{k,B}^\dagger d_{k,A}\right]$$

where

$$\Psi_k\equiv\frac{1}{\sqrt{N/2}}\sum_{\mathbf{r}}e^{-ik\cdot\mathbf{r}}\begin{pmatrix}c_{\mathbf{r},A}\\c_{\mathbf{r},B}\end{pmatrix}=\begin{pmatrix}d_{k,A}\\d_{k,B}\end{pmatrix} \tag{F.1}$$

$$U(k)=\frac{1}{2}\begin{pmatrix}0&is_{\mathbf{k}}\\-is_{\mathbf{k}}^*&0\end{pmatrix}\quad,\quad s_{\mathbf{k}}=2\left(J_xe^{ik\cdot(\mathbf{a}_2-\mathbf{a}_1)}+J_ye^{ik\cdot\mathbf{a}_2}+J_z\right)$$

Note that the index $\mathbf{r}$ above is the Bravais lattice point, and A,B represent the sublattice. If we further define a normal fermion operator,

$$\gamma_{\mathbf{r}}\equiv\frac{1}{2}\left(c_{\mathbf{r},A}+ic_{\mathbf{r},B}\right) \tag{F.2}$$

then its Fourier transform would be

$$\gamma_k = \frac{1}{\sqrt{N/2}} \sum_{\mathbf{r}} \gamma_{\mathbf{r}} e^{-ik\cdot\mathbf{r}} = \frac{1}{2}\left(d_{k,A} + i d_{k,B}\right)$$

In this way, the operator d can be expressed in terms of γ fermion,

$$\Psi_{\mathbf{k}} = \begin{pmatrix} d_{\mathbf{k}A} \\ d_{\mathbf{k}B} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma^\dagger_{-\mathbf{k}} \end{pmatrix} \tag{F.3}$$

Using the γ operators to rewrite the Hamiltonian gives

$$\hat{H} = \sum_{\mathbf{k}} \left[R_{\mathbf{k}} \cdot \left(\gamma^\dagger_{\mathbf{k}} \gamma_{\mathbf{k}} - \gamma_{-\mathbf{k}} \gamma^\dagger_{-\mathbf{k}} \right) - i I_{\mathbf{k}} \cdot \left(\gamma^\dagger_k \gamma^\dagger_{-\mathbf{k}} - \gamma_{-\mathbf{k}} \gamma_k \right) \right] = \sum_{\mathbf{k}} \phi^\dagger_k \mathbf{H}(k) \phi_k$$

where $R_{\mathbf{k}} \equiv \mathrm{Re}\left\{s_{\mathbf{k}}\right\}$ and $I_k \equiv \mathrm{Im}\left\{s_{\mathbf{k}}\right\}$, and

$$\mathbf{H}(k) \equiv \begin{pmatrix} R_{\mathbf{k}} & -iI_{\mathbf{k}} \\ iI_k & -R_{\mathbf{k}} \end{pmatrix} \quad , \quad \phi_k \equiv \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma^\dagger_{-\mathbf{k}} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \Psi_{\mathbf{k}}$$

Since

$$\mathbf{H}(k)^2 = \left(R^2_{\mathbf{k}} + I^2_k\right) \mathbb{1} = \left|s_{\mathbf{k}}\right|^2 \mathbb{1}$$

we can see that the eigenvalues are $\epsilon_k = \pm \left|s_{\mathbf{k}}\right|$, which is consistent with Kitaev's result (equation (32) in Ref.[8]). To diagonalize the matrix $\mathbf{H}$ we have to apply the Bogoliubov transformation,

$$\begin{aligned} U &= \frac{1}{\sqrt{2\left|s_{\mathbf{k}}\right|}} \begin{pmatrix} \mathrm{sgn}(I_k)\sqrt{\left|s_{\mathbf{k}}\right| + R_{\mathbf{k}}} & -i\sqrt{\left|s_{\mathbf{k}}\right| - R_{\mathbf{k}}} \\ i\sqrt{\left|s_{\mathbf{k}}\right| - R_{\mathbf{k}}} & -\mathrm{sgn}(I_k)\sqrt{\left|s_{\mathbf{k}}\right| + R_{\mathbf{k}}} \end{pmatrix} = \begin{pmatrix} \mathrm{sgn}(I_k)u_k & -iv_k \\ iv_k & -\mathrm{sgn}(I_k)u_k \end{pmatrix} \\ &= v_k \sigma^y + \mathrm{sgn}(I_k) u_k \sigma^z \end{aligned}$$

such that[4]

$$U \mathbf{H} U^\dagger = \begin{pmatrix} \left|s_{\mathbf{k}}\right| & 0 \\ 0 & -\left|s_{\mathbf{k}}\right| \end{pmatrix}$$

Therefore, the excitation that diagonalize the Hamiltonian is

$$\begin{pmatrix} a_k \\ a^\dagger_{-k} \end{pmatrix} = U \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma^\dagger_{-k} \end{pmatrix} = \begin{pmatrix} \mathrm{sgn}(I_k) u_k \gamma_{\mathbf{k}} - i v_k \gamma^\dagger_{-k} \\ i v_k \gamma_{\mathbf{k}} - \mathrm{sgn}(I_k) u_k \gamma^\dagger_{-k} \end{pmatrix} \tag{F.4}$$

As a result, the Kitaev ground state is the vacuum state of the Bogoliubov fermion, i.e., $a_k |M\rangle = 0$, which means $|M\rangle$ can be written as a BCS ground state just as we met in the BCS theory. The

[4] Those parameters in the matrix U have the following properties: **(i)** $u_k = u_{-k}$, **(ii)** $v_k = v_{-k}$, and **(iii)** $\mathrm{sgn}(I_k) = -\mathrm{sgn}(I_{-k})$

ground state can be written as the coherent state[5] of the γ-fermion pair,

$$\begin{aligned}|M\rangle &= A[g]\exp\left(\sum_{\mathbf{k}} g_k \gamma_k^\dagger \gamma_{-k}^\dagger\right)|\mathrm{vac}_\gamma\rangle \\ &= A[g]\exp\left(\sum_{\mathbf{r}\mathbf{r}'} g_{\mathbf{r}\mathbf{r}'} \gamma_{\mathbf{r}}^\dagger \gamma_{\mathbf{r}'}^\dagger\right)|\mathrm{vac}_\gamma\rangle \\ &= A[g]\left[\mathbb{1} + \sum_{\mathbf{r}\mathbf{r}'} g_{\mathbf{r}\mathbf{r}'}\gamma_{\mathbf{r}}^\dagger\gamma_{\mathbf{r}'}^\dagger + \frac{1}{2!}\sum_{\mathbf{r}\mathbf{r}'\kappa\lambda} g_{\mathbf{r}\mathbf{r}'}g_{\kappa\lambda}\gamma_{\mathbf{r}}^\dagger\gamma_{\mathbf{r}'}^\dagger\gamma_{\kappa}^\dagger\gamma_{\lambda}^\dagger + \cdots\right]|\mathrm{vac}_\gamma\rangle\end{aligned}$$

where $|\mathrm{vac}_\gamma\rangle$ is the vacuum state of the γ-fermions, and $g_{\mathbf{r}\mathbf{r}'} = \frac{1}{2N}\sum_{\mathbf{k}} g_k e^{-ik\cdot(\mathbf{r}-\mathbf{r}')}$.

F.1 With External Magnetic Field

For simplicity, we focus only on the isotropic limit [8]. In this situation, applying an external magnetic field $\mathbf{h} = (h_x, h_y, h_z)$, which is equivalent to adding a Zeeman perturbation $H' = -\sum_i \mathbf{h}\cdot\boldsymbol{\sigma}$, makes the Kitaev model equivalent to the Haldane honeycomb model [7] up to the third-order perturbation. The Hamiltonian under this approximation in the Majorana fermion representation reads

$$\begin{aligned}H = &\frac{1}{2}J\sum_{\mathbf{r}}\left(ic_{\mathbf{r}A}c_{\mathbf{r}+\hat{\mathbf{n}}_1,B} + ic_{\mathbf{r}A}c_{\mathbf{r}+\hat{\mathbf{n}}_2,B} + ic_{\mathbf{r}A}c_{\mathbf{r}B} + \mathrm{H.c.}\right) \\ &+ \frac{1}{2}\kappa\sum_{\mathbf{r}}\sum_{\mu=A,B}(-1)^\mu\left(-ic_{\mathbf{r}\mu}c_{\mathbf{r}+\hat{\mathbf{n}}_1,\mu} + ic_{\mathbf{r}\mu}c_{\mathbf{r}+\hat{\mathbf{n}}_2,\mu} - ic_{\mathbf{r}\mu}c_{\mathbf{r}+\hat{\mathbf{n}}_1-\hat{\mathbf{n}}_2,\mu} + \mathrm{H.c.}\right)\end{aligned}$$

Expand the Majorana fermions in terms of the momentum modes $c_{\mathbf{r}\mu} = \frac{1}{\sqrt{N/2}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} d_{\mathbf{k}\mu}$, the second term proportional to κ becomes

$$\kappa\sum_{\mathbf{k}}\sum_{\mu}(-1)^\mu\frac{1}{2}\left(-ie^{i\mathbf{k}\cdot\hat{\mathbf{n}}_1} + ie^{i\mathbf{k}\cdot\hat{\mathbf{n}}_2} - ie^{i\mathbf{k}\cdot(\hat{\mathbf{n}}_1-\hat{\mathbf{n}}_2)} + \mathrm{c.c.}\right)d_{\mathbf{k}\mu}^\dagger d_{\mathbf{k}\mu} \equiv \frac{1}{2}\sum_{\mathbf{k}}\Delta_{\mathbf{k}}\left(d_{\mathbf{k}A}^\dagger d_{\mathbf{k}A} - d_{\mathbf{k}B}^\dagger d_{\mathbf{k}B}\right)$$

where

$$\Delta_{\mathbf{k}} \equiv 2\kappa\left[\sin(\mathbf{k}\cdot\hat{\mathbf{n}}_1) - \sin(\mathbf{k}\cdot\hat{\mathbf{n}}_2) + \sin(\mathbf{k}\cdot(\hat{\mathbf{n}}_1-\hat{\mathbf{n}}_2))\right] \tag{F.6}$$

[5] The coefficient g_k can be represented in terms of u_k and v_k. As I have mentioned, the BCS-like ground state is expected to be the vacuum state of a_k fermions for all possible k, thus, we have

$$\left(\mathrm{sgn}(I_k)u_k\gamma_{\mathbf{k}} - iv_k\gamma_{-\mathbf{k}}^\dagger\right)\prod_k\left(\mathbb{1} + g_k\gamma_{\mathbf{k}}^\dagger\gamma_{-\mathbf{k}}^\dagger\right)|\mathrm{vac}_\gamma\rangle = 0$$

Based on this, we obtain that

$$\mathrm{sgn}(I_k)u_k g_k - iv_k = 0$$

therefore

$$g_k = \mathrm{sgn}(I_k)i\frac{v_k}{u_k} = \mathrm{sgn}(I_k)i\sqrt{\frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{|s_{\mathbf{k}}| + R_{\mathbf{k}}}} = i\frac{|s_{\mathbf{k}}| - R_{\mathbf{k}}}{I_k} \tag{F.5}$$

Converting the Hamiltonian into a spinor form defined by Eq. (F.1), we obtain

$$\begin{aligned} H &= \frac{1}{2}\sum_{\mathbf{k}} \Psi_{\mathbf{k}}^{\dagger} \begin{pmatrix} \Delta_{\mathbf{k}} & is_{\mathbf{k}} \\ -is_{\mathbf{k}}^{*} & -\Delta_{\mathbf{k}} \end{pmatrix} \Psi_{\mathbf{k}} \\ &= \frac{1}{2}\sum_{\mathbf{k}} \phi_{\mathbf{k}}^{\dagger} \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix} \begin{pmatrix} \Delta_{\mathbf{k}} & is_{\mathbf{k}} \\ -is_{\mathbf{k}}^{*} & -\Delta_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix} \phi_{\mathbf{k}} \\ &= \sum_{\mathbf{k}} \phi_{\mathbf{k}}^{\dagger} \begin{pmatrix} R_{\mathbf{k}} & \Delta_{\mathbf{k}} - iI_{\mathbf{k}} \\ \Delta_{\mathbf{k}} + iI_{\mathbf{k}} & -R_{\mathbf{k}} \end{pmatrix} \phi_{\mathbf{k}} \equiv \sum_{\mathbf{k}} \phi_{\mathbf{k}}^{\dagger} \mathbf{H}(\mathbf{k}) \phi_{\mathbf{k}} \end{aligned}$$

The BdG Hamiltonian represented in terms of Pauli matrices reads

$$\mathbf{H}(\mathbf{k}) = \Delta_{\mathbf{k}}\tau^{x} + I_{\mathbf{k}}\tau^{y} + R_{\mathbf{k}}\tau^{z} \tag{F.7}$$

$$\epsilon(\mathbf{k}) = \sqrt{|s_{\mathbf{k}}|^{2} + \Delta_{\mathbf{k}}^{2}}$$

The Bogoliubov transformation matrix is

$$\begin{aligned} U &= \frac{1}{\sqrt{2\epsilon(\mathbf{k})\left(\epsilon(\mathbf{k}) + R_{\mathbf{k}}\right)}} \begin{pmatrix} \epsilon(\mathbf{k}) + R_{\mathbf{k}} & \Delta_{\mathbf{k}} - iI_{\mathbf{k}} \\ \Delta_{\mathbf{k}} + iI_{\mathbf{k}} & -\epsilon(\mathbf{k}) - R_{\mathbf{k}} \end{pmatrix} \\ &= \frac{1}{\sqrt{2\epsilon(\mathbf{k})}} \begin{pmatrix} \sqrt{\epsilon(\mathbf{k}) + R_{\mathbf{k}}} & e^{-i\varphi_{\mathbf{k}}}\sqrt{\epsilon(\mathbf{k}) - R_{\mathbf{k}}} \\ e^{i\varphi_{\mathbf{k}}}\sqrt{\epsilon(\mathbf{k}) - R_{\mathbf{k}}} & -\sqrt{\epsilon(\mathbf{k}) + R_{\mathbf{k}}} \end{pmatrix} \\ &\equiv \begin{pmatrix} u_{\mathbf{k}} & e^{-i\varphi_{\mathbf{k}}}v_{\mathbf{k}} \\ e^{i\varphi_{\mathbf{k}}}v_{\mathbf{k}} & -u_{\mathbf{k}} \end{pmatrix} = U^{\dagger} = U^{-1} \end{aligned} \tag{F.8}$$

where $\varphi_{\mathbf{k}} \equiv \arg\left(\Delta_{\mathbf{k}} + iI_{\mathbf{k}}\right)$. Therefore, the elementary excitations that diagonalize the Hamiltonian are

$$\begin{pmatrix} a_{k} \\ a_{-k}^{\dagger} \end{pmatrix} = U \begin{pmatrix} \gamma_{\mathbf{k}} \\ \gamma_{-k}^{\dagger} \end{pmatrix} = \begin{pmatrix} u_{\mathbf{k}}\gamma_{\mathbf{k}} + e^{-i\varphi_{\mathbf{k}}}v_{k}\gamma_{-\mathbf{k}}^{\dagger} \\ e^{i\varphi_{\mathbf{k}}}v_{k}\gamma_{\mathbf{k}} + u_{\mathbf{k}}\gamma_{-\mathbf{k}}^{\dagger} \end{pmatrix} \tag{F.9}$$

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